DISASTER READY: A WEB-BASED APPLICATION FOR EVACUATION AND RESOURCE LOCATOR DURING CALAMITIES IN THE MUNICIPALITY OF LOS BAÑOS

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In Partial Fulfillment of the requirement for the Degree
Bachelor of Science in Information Technology
Specialized in Web and Mobile Applications Development

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The Researchers

DEDICATION

This study is wholeheartedly dedicated to the researchers' beloved families, whose unwavering love, support, and encouragement served as a constant source of motivation throughout the research journey. In particular, it is dedicated to their parents,

Mr. Norberto Jimeno and Mrs. Cecilia Jimeno,

Mr. Bonifacio Adorco and Mrs. Maricel Adorco,

and Mr. Dante Abrera and Mrs. Shirly Avila

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ABSTRACT

The study entitled Disaster Ready: A Web-Based Application for Evacuation and Resource Locator During Natural Calamities in the Municipality of Los Baños was developed to enhance disaster preparedness, communication, and emergency response within the local community. The system provides residents, barangay officials, and the municipal government with a centralized platform for accessing critical disaster-related information and locating evacuation centers during natural hazards such as typhoons, floods, landslides, earthquakes, and volcanic eruptions. The application integrates Geographic Information System (GIS) technology to support real-time visualization of evacuation centers, disaster-prone areas, and evacuation routing, enabling users to identify the safest routes during emergencies. Key features include real-time updates, emergency incident reporting, navigation to evacuation sites, and access to safety guidelines before, during, and after disasters. Developed using web technologies, the system ensures accessibility across multiple devices and browsers, supporting efficient coordination between residents and local authorities. An evaluation based on the ISO/IEC 25010 software quality model was conducted, covering functional suitability, performance efficiency, compatibility, usability, reliability, and security. Results indicated that the system achieved ratings ranging from Very Good to Excellent, demonstrating its effectiveness, reliability, and user acceptance. Overall, the Disaster Ready application provides an innovative and user-friendly solution that strengthens disaster management and community preparedness in the Municipality of Los Baños. Further enhancements are recommended, including offline functionality, real-time weather monitoring, and mobile optimization to ensure continuous operation during unstable conditions.

Keywords: *Disaster Management, Web-Based Application, Geographic Information System (GIS), Evacuation Routing, ISO/IEC 25010, Community Preparedness, Los Baños*

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DEFINITION OF TERMS

Calamity — In this study, calamity refers to natural disasters such as typhoons, floods, earthquakes, landslides, and volcanic eruptions affecting the Municipality of Los Baños.

Disaster Preparedness - — The process of ensuring that an organization or community is ready to respond effectively to emergencies and disasters.

Disaster Ready - (Web-Based Application) — A digital platform designed to provide disaster-related information and services accessible through the internet.

Evacuation Center - — A designated safe facility where people temporarily stay during and after a disaster to ensure their safety.

Flood-Prone Area - — A geographical location that is susceptible to flooding due to rainfall, topography, or poor drainage.

Landslide-Prone Area - — A location with a high likelihood of soil or rock movement, often caused by heavy rainfall or earthquakes.

Municipality of Los Baños - — A first-class municipality in the province of Laguna, Philippines, known for its research institutions and natural hot springs. The main area of focus for the “Disaster Ready” system where the disaster preparedness and response features are implemented.

Navigation System - — A feature in the application that guides residents to the nearest evacuation centers and safe routes during calamities.

Relief Distribution - Refers to the system feature that helps determine population counts and demographic details (seniors, adults, babies) for fair relief allocation per barangay.

Sub-Administrator - (Sub-Admin) Refers to barangay officials or representatives who input updates, population data, and reports within their assigned areas in the “Disaster Ready” application.

Web-Based Application - Refers to the online platform developed by the researchers that allows residents to access disaster information and services without installing separate software.

CHAPTER 1

INTRODUCTION

PROJECT CONTEXT

Disasters, both natural and man-made, pose a significant threat to communities, often leading to loss of lives and property. The Municipality of Los Baños, with a total population of 117,030, is particularly vulnerable to various natural calamities such as typhoons, floods, and earthquakes. Given this, there is a pressing need for a more efficient and technology-driven solution to enhance disaster preparedness and response. Traditional methods of emergency reporting and evacuation coordination often result in delays, misinformation, and inefficiencies, which can worsen the impact of disasters on affected communities.

According to the United Nations Office for Disaster Risk Reduction (UNDRR, 2015), a disaster is a serious disruption of the functioning of a community or society caused by hazardous events, resulting in widespread human, material, economic, or environmental losses that exceed the affected community's capacity to cope using its own resources. Disasters may be natural or human-induced and can significantly impact not only human lives but also animals, ecosystems, infrastructure, and economic stability. Natural disasters such as typhoons, floods, earthquakes, landslides, and volcanic eruptions often cause extensive damage to buildings, disrupt livelihoods, and slow national and local economic growth. These events highlight the importance of preparedness, effective response systems, and reliable

information dissemination to reduce risks and mitigate long-term impacts on communities and development.

Evacuation is fundamental to saving lives during natural disasters. People in the Philippines may be endangered if they do not know the proper way to evacuate during disasters, especially during stampedes in Local Government Units (Atacador, Iglesias & Beron, 2019). Traffic congestion, especially during typhoons, can trap people. The average number of typhoons per year in the Philippines makes it difficult to handle. The primary function of evacuation is to prevent damage and save people in safe zones. Early warnings and emergency announcements can reduce fatalities (Yan, Liu & Song, 2018).

Comparing two different areas that experienced typhoons, the fewer fatalities that followed the authorities' instructions immediately resulted in less extensive damage. The cost of responding to hundreds and millions of dollars in lost human lives due to their geographic location is frequently occurring in the Philippines (Walch, 2018). Natural disasters in the Philippines include typhoons, volcanic eruptions, earthquakes, floods, and landslides. The Philippines experiences significant exposure to storms and land in the country. Averaging 18 to 20 significant tropical cyclones per year, the National Disaster Risk Reduction and Management Council (NDRRMC) is responsible for gathering disaster-related information. It is the largest platform that handles Disaster Risk Management in the country. Disaster preparedness in the affected areas is periodically reviewed and updated. A typhoon in 2021, named "ODETTE", accumulated the record of the 15th super typhoon in the NDRRMC that hit the country. 11 regions were affected; 7,846,542 people

were injured, 52 went missing, and 405 died in December 2021 (NDRRMC, 2022). A mobile application provides the shortest path to the safest place or evacuation centers during disasters, ensuring the safety of residents in various barangays of Los Baños, Laguna.

To address these concerns, the development of a web-based application called "Disaster Ready" serves as an innovative solution that aligns with the municipality's goal of improving disaster preparedness and response strategies. By integrating web-based platforms, this system would provide real-time updates on evacuation centers, resource availability, and emergencies, ensuring that residents and authorities are well-equipped to handle natural disasters effectively.

PROJECT OBJECTIVES

The main objective of this study is to develop a Web-based Application for evacuation and Resource Locator that would raise awareness among the residents who are affected by natural calamities in Los Baños. This project aims to:

1. Develop a Web-based Application to inform residents about disaster preparedness and partner with local officials for disaster information.

2. Provide real-time information for available evacuation centers and alternative routes using GIS technology, especially during live system operations.

3. Conduct system testing using the ISO25010 models to ensure system quality, reliability, and user acceptance.

PROJECT PURPOSE

The purpose of the study is to guide the residents of Los Baños in case of disaster. The proponent believes that this study can be beneficial to the following:

Disaster Victims. This study's results would primarily benefit Los Baños, Laguna residents by improving their understanding of and preparedness for natural disasters. The increasing frequency and severity of typhoons, earthquakes, and the potential for a major earthquake ("The Big One") highlight the critical need for this research. By increasing awareness and providing access to safety information, this research aims to reduce the risks to lives during disasters significantly. Knowing where to find safety quickly can make a crucial difference. The study’s implementation assists residents in determining the shortest path from their location to the evacuation sites. The implementation was completed as soon as possible through research and hard work to serve as a guide in ensuring the safety of the residents.

Municipal Disaster Risk Reduction and Management. This application can significantly benefit a Municipal Disaster Risk Reduction Management program by supporting a quick and efficient response, as it quickly identifies individuals in need of assistance during a disaster. This can help save precious time that can be critical in life-threatening situations. It can also allocate resources more effectively, allowing areas that require immediate attention to be addressed and enabling the team’s full potential to be utilized. This can also enhance the team’s communication, as updates can be

accessed immediately and people can use it to ask for help and share their location.

Barangays. Anyone with a web or mobile device can use the application. It helps ensure a quick and efficient response to disasters, better coordination among authorities, and improved communication between authorities and those in need.

Researchers. This project enhances their technical skills and web-based application development while addressing real-world issues in disaster preparedness. It strengthens their problem-solving, critical thinking, and project management abilities, and contributes to their academic growth in Disaster Risk Reduction and Technology.

Future Researchers. Future researchers can build upon this system by enhancing its real-time capabilities, integrating more advanced Geographic Information System (GIS) features, or expanding the application to cover other municipalities and regions. They may also explore the incorporation of machine learning algorithms to predict the impact of disasters or automate resource allocation based on real-time data.

Scope and Limitations of the Study

This project is a web-based navigation system for the municipality of Los Baños. It allows users to browse and check routes to evacuation centers, receive the latest updates about the community and calamities, and submit an incident report when immediate rescue is needed during emergencies.

The app has limitations; it requires a stable internet connection to function effectively, which may be disrupted during disasters. Its coverage is limited to the Municipality of Los Baños, making it unavailable to nearby towns.

The accuracy of information also depends heavily on timely updates from local government officials. Furthermore, once a Low-Pressure Area (LPA) enters or heavy rainfall begins, residents are advised to save important contact information of LGUs in advance. This ensures they can still reach out for help through calls or text messages, even without load for mobile data connection or when Wi-Fi is unavailable.

The project consists of the following modules:

Home page. The home page serves as the primary access point for both users. Upon prompting in, users are greeted with updates and news that include road advisories, weather warnings, and community notices, ensuring they stay informed before, during, and after any natural disaster. The dynamic home page is tailored specifically to the Municipality of Los Baños, thereby promoting awareness and proactive action among residents.

Donate Now. This module empowers community members to contribute to disaster relief efforts. It supports multiple donation modes, including e-wallet transfers and bank deposits. By integrating this module, the app bridges the gap between willing donors and those in need, especially in times of calamities, facilitating quicker mobilization of relief efforts across the municipality.

Report Incident Module. Designed to streamline communication during emergencies, this module enables residents to quickly send reports through the app. Upon receiving the report, the admin reviews and validates the information before publishing it as news or official announcements. This ensures that only verified data circulates within the system, supporting

accurate decision-making and timely interventions by emergency responders and local government units.

Safety Tips. This educational module is curated to equip residents with essential knowledge about the various types of disasters they may face in Los Baños, including typhoons, earthquakes, floods, landslides, and volcanic eruptions. It offers practical safety guidelines organized by disaster phases (before, during, and after). This feature promotes disaster literacy and individual preparedness, reducing potential harm through proactive behavior.

Emergency Resources. This module provides real-time access to a directory of available emergency resources per barangay, including designated evacuation centers, medical facilities, and local emergency hotlines. It ensures that during a calamity, residents can swiftly identify and reach out to critical support channels, thereby minimizing confusion and enhancing coordinated community responses.

Map. The Map feature displays important real-time markers such as pharmacies, hospitals, fallen trees, road blockages, landslide-prone areas, and flood-affected zones, providing users with a complete situational overview during emergencies. This feature is crucial for ensuring that residents make well-informed decisions in rapidly changing emergency scenarios.

Account Setting. This module enables all user types to manage their profile information, including updating personal details, changing passwords, and setting communication preferences. Maintaining accurate and secure account data is vital for identity verification during emergency interactions and data logging.

Notification. In urgent situations, both admin levels immediately receive notifications through the system as soon as a resident submits an emergency report. The system automatically sends these alerts in real time, ensuring that administrators are promptly informed without needing to manually check the dashboard. Once an admin responds, the system also automatically notifies the resident that responders are already on the way, preventing any uncertainty or long waiting without updates. After the concern has been fully addressed, the issue must be marked as completely responded to maintain accurate tracking of incidents. This automated notification process enhances coordination by ensuring that both responders and residents receive timely updates, leading to quicker action, faster resolutions, and more efficient communication between the community and local authorities.

Login and Logout. The Login and Logout Module is designed exclusively for Sub Admin and Super Admin accounts, ensuring secure access to the administrative functions of the Disaster Ready system. When a Sub Admin wants to create an account, they cannot access the system immediately. Instead, they must submit an account request through the registration form, which is then forwarded to the Super Admin for verification. This validation step allows the Super Admin to confirm whether the applicant is an authorized barangay representative before approving or rejecting the request. This process prevents unauthorized individuals from gaining administrative privileges, ensuring that only legitimate barangay officials can manage disaster-related data. Once approved, the Sub Admin can log in using their verified credentials and access modules such as Dashboard, Evacuation Centers, Incident Reports, Updates and News, and Barangay Contact Information,

Disaster Risk Mapping, Account Setting, depending on their assigned permissions. Both Sub Admins and Super Admins can securely log out after managing tasks, ensuring that sensitive information remains protected.

Dashboard. The Dashboard module provides a centralized interface where admins can monitor and analyze key metrics, such as the total population in evacuation centers, recent reports, and current disaster alerts. It offers a real-time overview of the system's operations and is essential for informed planning and coordinated response strategies.

Evacuation Center. This module allows administrators to create, update, and manage information about evacuation centers, including their capacities, locations, availability, and current status. Accurate data helps avoid overcrowding and ensures resources are evenly distributed during emergencies, supporting efficient evacuation planning and coordination. Users can also view the nearest evacuation centers based on their current location and access a guided route showing step-by-step instructions on how to reach the selected center. This feature ensures safer, faster, and more informed decision-making during disaster situations.

Barangay Contact Information. Admins can maintain updated contact details for each barangay disaster response team. This module ensures that residents can access reliable contact points in their respective areas, especially during critical times when local coordination is necessary for rescue.

Relief Goods. This feature enables the Super Admin to track the number of families in each barangay, facilitating the efficient distribution of

relief goods. The system ensures that no community is left behind and that the resources are dispatched equitably based on actual needs, population size, and severity of impact.

Disaster Risk Mapping. Admins can regularly update the visual map integrated into the app, which displays disaster-prone zones, ongoing hazards, and routes. This visual intelligence tool is vital for planning evacuations, guiding users to safety, and minimizing exposure to high-risk areas during natural disasters.

Updates and News. This module serves as the main communication hub for delivering verified information to the community. Both Admins can publish important news, advisories, and announcements about ongoing or potential disasters. It keeps residents informed about weather updates, road closures, and evacuation notices in real time, helping prevent misinformation and panic. By ensuring accuracy and credibility, this module strengthens coordination between authorities and residents, promoting transparency, preparedness, and trust within the community.

User Management. This module allows the Super Admin to manage and monitor all registered users in the system, ensuring proper authorization and secure access. It enables deactivating and blocking, managing permissions to prevent misuse. By maintaining strict control over user roles, this feature upholds system security, data integrity, and efficient coordination between Super admin and sub-admins within the Disaster Ready application.

CHAPTER 2

THEORETICAL FRAMEWORK

Review of Related Literature, Studies, and Systems

This chapter presents foreign and local literature that emphasizes the development of web-based applications that can provide information during a disaster.

Disasters

According to the United Nations Office for Disaster Risk Reduction (UNDRR, 2015), a disaster is a serious disruption of the functioning of a community or society caused by hazardous events, resulting in widespread human, material, economic, or environmental losses that exceed the affected community's capacity to cope using its own resources. Disasters may be natural or human-induced and can significantly impact not only human lives but also animals, ecosystems, infrastructure, and economic stability. Natural disasters such as typhoons, floods, earthquakes, landslides, and volcanic eruptions often cause extensive damage to buildings, disrupt livelihoods, and slow national and local economic growth. These events highlight the importance of preparedness, effective response systems, and reliable information dissemination to reduce risks and mitigate long-term impacts on communities and development.

Evacuation centers

Evacuation centers are temporary homes for people affected by a disaster. They are the safe zones in the community where affected people can

stay. Evacuation centers provide basic needs such as food, clothes, and water. Preparing emergency kits and needs is a routine for risk reduction workers to use in places that can experience disasters such as typhoons, fire accidents, volcanic eruptions, and earthquakes. (Onorati, et al. 2014).

The municipality of Los Baños, Laguna, is prone to disaster, specifically typhoons, because Laguna Lake and creeks surround it. (Catapang, 2016) Most of the time, barangays experience floods named Tadlac. Lalakay, Baybayin, Mayondon, Malinta, and Bambang. (Grefaldo, 2016) Local government officials are informing the residents where to go during the disaster. There is an evacuation center allotted to every barangay. Some are schools and centers; they have Los Baños National High School, Lopez Elementary School, and a multipurpose center in the city market. (MDRRMO, 2016)

Knowing the capacity of the safe zone is a requirement for LGUs. Sorting residents beforehand is a good practice in case of an emergency. It is a way to prevent people from jams and stampedes. (GovPilot, 2021).

Evacuation Plan and Routes

The Evacuation Plan is sifted through for a given travel time and a particular flooding area; residents of coastal areas must flee to a safe location as soon as a disaster occurs, and the prime evacuation period is crucial for them. It is a method of saving people’s lives. (Putra & Rohendi, 2017). In an emergency, a designated evacuation route is created to move people from buildings and locations as quickly as possible. These pathways from a starting

point (office, building, street, mountain, etc.) to a safe location, typically outside, must be continuous and unrestricted. (Yoo & Choi, 2019).

In communication, smartphones are what we need during emergencies. It is a big help to keep it closer to us because there is much help, especially nowadays that we live in a technological era. The source of information during the disaster. It can give information on how they would evacuate in the area, check what available ways or routes to go in the nearest evacuation to their current location, and communicate to authorities by choosing safe and needed rescue in the mobile application (Putra & Rohendi, 2017).

Using the GIS (Geographical Information System), you can know the area's elevation and information about the area's country, the road access, and the percentage of the affected area. The system can also assess what evacuation center is suitable for their location and analyze the existing data in the evacuation centers in the city. (Mustafa, et al., 2015).

Web-based Application

Web-based applications have become a vital tool in delivering services that require real-time data access, scalability, and accessibility across various devices. These systems, which run on browsers without the need for installation, are widely used in domains such as health, education, e-commerce, and disaster management (Alshamrani & Bahattab, 2021).

One of the major advantages of web-based systems is their accessibility. As noted by Dahiya and Bhatia (2020), users can interact with the application anytime and anywhere, provided there is internet connectivity. This

characteristic makes them especially useful in emergencies where timely access to information is crucial. Web-based platforms have been successfully utilized in early warning systems and crisis mapping tools, enabling authorities to disseminate information quickly and coordinate response efforts (Meissen et al., 2020).

In disaster management, web applications are often integrated with geolocation features, such as Google Maps API, to provide users with real-time locations of evacuation centers, affected zones, and relief distribution points. Studies by Nguyen et al. (2019) highlight how such integrations improve situational awareness and aid in faster decision-making for both citizens and responders.

User-centered design is also emphasized in modern web applications. According to Nielsen (2020), the usability of web-based platforms significantly affects user adoption, especially in high-stress contexts like natural calamities. Features like simplified reporting forms, clear visual cues, and mobile responsiveness are essential in ensuring that the application can be effectively used by people of varying tech literacy levels. Security and data privacy are also key concerns. With systems handling sensitive data such as user locations and personal information, developers are encouraged to implement secure authentication methods and encrypted data handling processes (Owens et al., 2021).

Web-Based Geographic Information Systems

Web-Based Geographic Information Systems (Web GIS) have become essential tools for spatial data visualization, analysis, and decision-making

across various sectors, including disaster management. Unlike traditional GIS, Web GIS allows users to interact with geospatial data through web browsers, offering greater accessibility, scalability, and real-time data sharing (Zhang et al., 2018).

In the field of disaster response, Web GIS plays a crucial role by enabling users to visualize hazard-prone areas, locate evacuation centers, and track the distribution of resources during emergencies. According to Goodchild and Glennon (2010), the integration of GIS into web applications enhances situational awareness and supports rapid decision-making in dynamic environments. These systems allow local governments to make data-driven decisions and provide citizens with real-time updates on disaster status and safe zones.

The use of Web GIS in emergency scenarios has been demonstrated in several projects. For example, the Sahana Disaster Management System incorporates GIS features for mapping shelters and affected areas (Currion et al., 2007). Similarly, the GeoNode platform provides open-source tools for sharing geospatial data, which can be critical during natural calamities such as floods or typhoons (Hjelmager et al., 2020).

Modern Web GIS solutions commonly use technologies such as the Google Maps API, Leaflet.js, or ArcGIS Online to deliver interactive maps. These tools enable dynamic layering of data, real-time updates, and location-based services, all of which enhance user engagement and system utility (Esri, 2019). The integration of geospatial functions with mobile and web

technologies makes these systems suitable for municipalities seeking to improve disaster preparedness and citizen safety.

User accessibility is another advantage. Web GIS systems are typically device-agnostic and mobile-friendly, allowing residents to access location-based services through smartphones or computers. This improves public reach and ensures that even users in remote areas can benefit from timely, map-based information (Chen et al., 2021).

Zones

The purpose of an evacuation is to move as few people as possible in the shortest amount of time to safety. This goal can be best met with zone-based evacuation preparations. They lessen resource demands, facilitate re-entry, and hasten the rehabilitation process. A zone-based approach necessitates collaboration between emergency management and the entire community to comprehend and coordinate evacuation and shelter-in-place measures, as well as make educated decisions based on the proper transportation models.

Evacuation Clearance

Time The expected amount of time needed to safely evacuate individuals from the time an evacuation order is issued until either the last evacuee departs the remaining population is either forced to shelter in place because operations have been suspended as the hazard starts to affect the area given multiple considerations such as the type of danger or threat, the

degree of encounter of the situation, characteristics of populations of the area at the time, and public behavior.

Routing Optimization

The goal of routing optimization problems is to find effective routes between network nodes, such as simple routes, closed routes, tours, or walks. These issues and their solutions have become crucial components for resolving a variety of real-world choices in applications as diverse as logistics, transportation, computer networking, and internet routing, among others. The existence of uncertainty in the networks (such as the presence or absence of some of the nodes) in many of these routing applications, especially those imposing deadlines on when to visit nodes, is a critical issue that must be explicitly taken into consideration if one hopes to offer solutions of usefulness to the end users.

History of the Global Positioning System (GPS)

To enhance navigation and aviation safety, President Ronald Reagan approved the use of Navstar (later known as GPS) by civilian commercial airplanes in 1983. The first step toward approving civilian usage was the decision to grant open access to GPS data to businesses outside of the U.S. military. The Magellan Corporation's MagellanNAV 1000, a handheld GPS device that cost $3,000, weighed 1.5 pounds, had a battery life of only a few hours, and was commercially available by 1989. The PS technology developed further in the 1980s and 1990s. The first operational GPS Block II satellite was launched on a Delta II rocket in 1989 after the production and development phase had started in 1985. The government also began to worry

that American rivals might use the GPS and its data against the country. In response, the Defense Department decided to modify the system to be less accurate in applications other than military ones (known as "selective availability") to thwart any use by enemies. GPS's influence on everyday life grew as its coverage continued to grow until it reached its maximum operational capacity. When Benefon debuted the Benefon Esc in 1999, GPS technology made its debut in a mobile phone, a phone with GPS that would point the way for more. Additionally, GPS technology started to appear in vehicles. The government approved proposals in 2000 to add three more GPS signals for non-military use. More crucially, the US government terminated the "selective availability" initiative, which reduced the accuracy of civilian GPS users. By the year 2000, it was clear that the system needed to be upgraded to fulfill the fast-expanding military and civilian applications. A mechanism for adding new signals to unlaunched satellites was created; however, satisfying the upgraded requirements would necessitate a new program known as GPS III. By 2005, the Block II satellites were available in five different configurations, each with its own set of capabilities: original IIs, IIAs, and IIRs were already in orbit; eight updated IIR-Ms began to be launched, and 12 modernized IIFs were in full development. The first upgraded IIF was released in 2010, with the remainder following suit by 2017. The remaining six satellites would be launched by 2023. GPS III Follow-on (GPS IIIF) satellites are now being developed to complete a full constellation of updated GPS satellites. The influence of space-based navigation technology on our environment is still growing. Not only is the direct PNT data valuable, but so is the integration of that data with other data for maps, weather, and synchronized/crowd-

sourced data. According to reports, GPS satellites have contributed approximately $1.4 trillion in economic gains since the 1980s. PNT timing is critical for the operation of the data networks and financial systems. They utilize it to do scientific research on earthquakes, volcanoes, and the movement of tectonic plates. Space-based navigation is utilized to enhance building and optimize farming practices, such as water and pesticide application (Aerospace 2020).

Global Positioning System (GPS)

GPS is an abbreviation for Global Positioning System. GPS (Global Positioning System) is crucial in mobile applications for various reasons. It is a satellite navigation system used to pinpoint a device's exact location and time on Earth. The system employs an orbiting satellite network that sends signals to GPS receivers on the ground. These receivers use satellite signals to calculate their precise location (latitude, longitude, and altitude) and time. GPS is commonly used for navigation and tracking and may be found in various devices such as smartphones, automobiles, and boats. Many businesses, including transportation, agriculture, surveying, and the military, also employ it. GPS can be used to pinpoint a user's exact location, allowing it to deliver location-based information such as nearby businesses, sites of interest, and driving or walking directions. GPS can be used as a turn-by-turn navigation tool, delivering real-time updates and traffic information to assist users in reaching their destination as quickly as possible. The software can deliver customized recommendations, notifications, and offers based on the user's home and work location. It can measure users' activity and mobility, providing data on distance traveled, pace, and calories burned. It enables fleet

management and logistics optimization by tracking and monitoring the real-time position of vehicles, cargo, or personnel using smartphones.

Leaflet Map

Leaflet is an open-source JavaScript library designed for interactive maps, widely recognized for its lightweight nature, high performance, and user-friendly interface. Developed by Vladimir Agafonkin in 2011, Leaflet has become one of the most popular mapping libraries in web development due to its simplicity and flexibility. According to Agafonkin (2011), the library focuses on providing the essential functions needed to create dynamic, mobile-friendly maps that can be easily integrated into web applications. Leaflet supports multiple map tile providers, including OpenStreetMap, Mapbox, and Esri, making it a versatile option for developers seeking customizable mapping solutions.

Zeng and colleagues (2020) emphasized that the use of web-based mapping tools like Leaflet enhances real-time visualization of spatial data, particularly in disaster management systems. By integrating Leaflet with geolocation APIs and real-time databases, users can track incidents, visualize evacuation routes, and identify hazard zones efficiently. Its compatibility with REST APIs and plugins allows developers to extend its functionalities — such as adding markers, pop-ups, or heat maps — to visualize data intuitively.

A study by Johnson (2019) highlights that Leaflet's performance on mobile devices remains one of its strongest advantages compared to other mapping libraries such as Google Maps API and OpenLayers. Its small file

size (approximately 38 KB) and modular architecture allow faster rendering of maps even on limited network connections, which is crucial in emergency or disaster scenarios where stable internet connectivity might be unavailable.

In the Philippines, various researchers and developers have utilized Leaflet in building web and mobile-based geographic information systems (GIS). De Guzman and Santos (2021) developed a Barangay Mapping and Information System using Leaflet.js to visualize demographic and disaster-related data within local communities. Their study showed that Leaflet's open-source nature provides cost-effective and efficient mapping capabilities for local government units that lack access to premium mapping APIs.

Similarly, Dela Cruz et al. (2022) created a Flood Monitoring and Evacuation Route System for Marikina City, integrating Leaflet maps with GPS tracking and rainfall sensor data. Their findings indicated that using Leaflet enhanced data visualization and improved the accessibility of evacuation center information, allowing residents to view safe routes and flood-prone areas in real time.

A study by Liang et al. (2020) on emergency management applications demonstrated that integrating Leaflet with cloud-based geospatial databases enables the delivery of real-time situational awareness during disasters. Their prototype system visualized evacuation shelters, transportation routes, and affected populations dynamically, helping authorities make data-driven decisions faster.

Another research by Ahmed and Al-Shammari (2019) used Leaflet in developing a web-based risk mapping tool for earthquake risk assessment. The study concluded that Leaflet's simple API and ability to render interactive map layers significantly improve the usability of GIS applications in disaster response and preparedness planning.

In the local context, research by Villanueva and Ramos (2023) utilized Leaflet for a "Community-Based Disaster Preparedness and Evacuation System" in Laguna. The study showed that Leaflet's integration with open-source datasets from OpenStreetMap provided accurate visual representations of barangays, evacuation centers, and hazard-prone areas. The researchers concluded that using Leaflet significantly improved public accessibility to spatial disaster information and strengthened coordination among local emergency units.

Moreover, Bautista (2022) developed a mobile-based "Evacuation and Relief Management Application" using Leaflet for visualizing affected zones and safe routes. Results showed that Leaflet's customizable markers and routing plugins enhanced the user's experience by providing an intuitive interface for locating evacuation centers and tracking donations during calamities.

Typhoon

A typhoon is a highly organized storm system that forms from cyclone particles. It matures by absorbing significant amounts of water vapor that condense at higher altitudes from the warm tropical oceans. As the typhoon moves across the Pacific Ocean, its primary source of energy is the heat of

condensation. Typhoons pose severe risks to human life and urban development due to their mighty destructive power and additional massive storm hazards, such as storm surge flooding and excessive rainfall. As a result of global warming, powerful typhoons have happened more frequently, and marine disasters have also increased (Anduaga, 2019). About 20 typhoons, or 25% of all typhoons worldwide, land in the Philippine Area of Responsibility each year. Typhoons primarily strike during the rainy season (July until September) (Santos, 2020).

Floods

A flood is a natural disaster that occurs when excess water submerges normally dry land, resulting in serious damage to human lives, infrastructure, animals, and plants. It is commonly caused by prolonged heavy rainfall, rapid snowmelt, or storm surges from tropical cyclones or tsunamis, particularly in coastal and riverine areas (WHO, 2017). Floods may develop gradually, allowing communities time to prepare or evacuate, or they may occur suddenly with little warning, such as during dam failures or flash floods. As one of the most frequent and destructive natural disasters worldwide, floods can destroy buildings, erode soil, contaminate water sources, and displace communities. According to Boudreau et al. (2022), floodwaters often carry hazardous materials and sewage, increasing the risk of disease outbreaks such as cholera and malaria, as seen in past flood events where poor sanitation and stagnant water worsened public health conditions.

Typhoon Kristine (international name: Trami), which struck the Philippines in late October 2024, significantly affected Southern Luzon,

including the municipality of Los Baños, Laguna. According to the Philippine Atmospheric, Geophysical and Astronomical Services Administration (PAGASA, 2024), Typhoon Kristine brought sustained winds of up to 85 km/h and heavy rainfall that led to flooding and landslides in several municipalities in Laguna province. The prolonged rainfall caused water levels in nearby rivers, such as the Molawin and Dampalit Rivers, to rise beyond critical levels, inundating low-lying barangays in Los Baños.

The Municipal Disaster Risk Reduction and Management Office (MDRRMO) of Los Baños (2024) reported that numerous barangays, including Anos, Batong Malake, and Mayondon, experienced flooding that displaced residents and damaged properties. The local government activated evacuation centers and implemented pre-emptive evacuations to ensure the safety of residents, but challenges in communication and coordination were observed. This incident underscored the importance of having an effective and technology-driven disaster management system that could provide real-time updates and facilitate quick response actions.

In a report by the Philippine News Agency (2024), Typhoon Kristine caused power outages, road blockages due to fallen trees, and temporary suspension of classes and local business operations in Laguna. The disaster revealed the vulnerability of Los Baños' critical infrastructure to typhoon-related hazards and emphasized the need for sustainable and climate-resilient urban planning. Furthermore, the flooding disrupted agricultural activities in nearby barangays, affecting small-scale farmers who rely on crops such as rice and vegetables for their livelihood.

The Department of Science and Technology (DOST, 2024) also noted that Los Baños, being situated near Laguna de Bay and the foothills of Mt. Makiling, is prone to hydro-meteorological hazards. The agency emphasized the value of geospatial and mapping technologies, such as the use of Leaflet maps and GIS-based platforms**,** to monitor hazard-prone areas and improve evacuation planning. Integrating such digital tools into local disaster response systems can significantly enhance preparedness, which aligns with the objectives of the Disaster Ready web application project.

Fire Incidents

A fire incident involves smoke, heat, and flames, causing property damage to multiple buildings and infrastructure. Due to the harm it does to infrastructure and buildings. During a fire, poor air quality, loss of life, and destruction of property occur, and damaged areas are more prone to floods, debris flows, landslides, and poor water quality in the aftermath. Longer fire seasons have resulted from climate change, as vegetation dries up earlier and sustained high temperatures allow fires to burn longer. Fire can have catastrophic consequences for human society.

Volcanic Eruption

A Volcanic eruption is the release of gas, ash, molten materials, or hot water into the atmosphere or onto the Earth's surface from a volcano or other openings in the Earth's surface. In the past 20 years, the Philippines has experienced six volcanic eruptions. The volcanoes' names are Bulusan, Mayon, and Taal. (World Data, 2020). The Taal Volcano Eruption is the one that affects the City of Los Banos. It happened last January 2020. The whole city experienced ash fall. The roads, the roofs of houses, and everything else

are gray. After the eruption, the LGUs cleaned the roads using fire trucks, and the residents started cleaning their environment and roofs (Batong Malake BPSO, 2020).

Earthquake

An earthquake is a sudden shaking of the ground caused by seismic waves passing through the Earth's rocks. Earthquakes occur most frequently along geologic faults, narrow zones where rock masses shift concerning one another. The world's primary fault lines run along the edges of the massive tectonic plates that make up the Earth's crust. Every year, around 50,000 earthquakes significant enough to be felt without the use of instruments occur around the world. Over 100 of these are large enough to cause significant harm if their centers are near places of habitation. Large earthquakes happen around once a year on average. Over the centuries, they have been responsible for millions of deaths and an incalculable amount of property damage. The Circum-Pacific Belt is the most important earthquake belt, affecting numerous populous coastal regions across the Pacific Ocean. It is estimated that 80 percent of the energy released in earthquakes today comes from those with epicenters in this band. Seismic activity is not uniform over the belt, and there are several branches at various sites. Because the Circum-Pacific Belt is associated with volcanic activity in several places, it has been nicknamed the "Pacific Ring of Fire." Earthquakes are triggered by the sudden release of energy within a specific section of the Earth's rocks. Elastic strain, gravity, chemical reactions, and even the motion of huge masses can all produce energy. The release of elastic strain is the most important cause of all because it is the only type of energy that can be stored in large quantities

in the Earth to create major upheavals. Earthquakes caused by this type of energy release are known as tectonic earthquakes (Bolt, 2022).

Table 1. Literature Matrix

Author	Year	Title	Findings
Rebecca Thompson, Dana Griffin, Roxane Silver	2017	Evacuation from Natural Disaster: A Systematic Review of the Literature. Society of Risk Analysis	Risk perception consistently predicts evacuation; demographic factors, prior evacuation behavior, and having an evacuation plan are less frequently examined. prospective, longitudinal studies are challenging but needed to improve inferences; current knowledge on evacuation behavior can inform more effective warnings and procedures.
Esposito Amideo, et al.	2019	Optimising shelter location and evacuation routing operations. European Journal of Operational Research	Optimization addresses challenges in disaster management related to shelter location and evacuation routing; historically tackled independently, but recently combined models have limitations and lack inclusivity; practical application in real-world situations remains limited.
Gesine Honfiger, et al.	2014	Human factors in evacuation simulation, planning, and guidance	Studies on evacuation have gained increased attention in human factors and psychology;

			previously, humans were portrayed as uniform, without considering individual emotions, motivations, or physical requirements; the focus was primarily on physical attributes and spatial requirements.
Hector Lim Jr. et al.	2017	A Review of Recent Studies on Flood Evacuation Planning. Eastern Asia of Society Transportation Study	While flooding is a natural phenomenon essential for biodiversity, human activities and climate change have resulted in more frequent and severe floods. To avoid the negative consequences of flooding, preparedness measures such as evacuation are increasingly necessary. This review focuses on recent advancements in evacuation research from the perspectives of behavioral science, risk analysis, and transportation modelling.
Hiromi Kawasaki, et al.	2020	Teacher-parent cooperation disaster preparation when schools become evacuation centers. International Journal of Disaster Risk Reduction	Since the Great East Japan Earthquake, disaster preparedness initiatives, including strengthening disaster education and management in schools, have been implemented. However, teachers were not able to focus on regular

			education due to their involvement in post-disaster management when schools were used as evacuation centers.
Jennifer Collins et al.	2017	Evacuation Behavior Measured During Evacuation Order	The study found that evacuees had denser and diverse relationships, but there was no significant relationship between the perceived dependability of social connections and evacuation decisions.

Synthesis and Relevance of the Study

People affected by disasters temporarily reside in evacuation centers because they offer safety. Evacuation centers must provide necessities like food, clothing, and water. Risk reduction professionals regularly prepare emergency kits and supplies for disasters like typhoons, fire accidents, volcanic eruptions, and earthquakes (Onorati et al., 2014). Los Baños Laguna is prone to natural disasters like floods due to its surrounding lakes and creeks. Many barangays frequently experience floods. Local government officials inform residents of assigned evacuation centers during disasters (MDDRMO, 2016).

The evacuation plan should be carefully examined for specific travel times and flooding areas. Coastal residents must leave immediately after a disaster; preserving the critical evacuation window is essential (Putra & Rohendi, 2017). Designated evacuation routes must quickly move people out of buildings to safe areas (Yoo & Choi, 2019).

A typhoon is a highly organized storm system that forms from cyclone particles and matures by absorbing water vapor from warm tropical oceans. The condensation at higher altitudes, with the Pacific Ocean as the primary source of energy, leads to the heat of condensation. Typhoons pose severe risks due to their destructive power and additional massive storm hazards like storm surges and excessive rainfall (WHO, 2017).

Floods are a common disaster caused by excess water submerging dry land. Fire incidents involve smoke, heat, and flames, causing property damage. Volcanic eruptions release gas, ash, molten materials, or hot water into the atmosphere (Res, 2019). Earthquakes are uncontrollable and dangerous natural hazards that cause physical, psychological, and social damage (Wald, 2010).

Conceptual Framework

The figure below illustrates the overall flow and interaction within the Disaster Ready, a web-based application, which aims to provide timely evacuation and resource information to residents of Los Baños during natural disasters.

The process begins with the User, who can submit emergency reports, and access donation features through the application. Once the user performs these actions, the data is sent to the Web-based application backend, which is powered by PHP and connected to a MySQL database. This backend is responsible for processing data, validating reports, and managing user input, ensuring accuracy and reliability. The application is managed by two types of administrators: Sub-Admin and Super-Admin. They handle different functions such as data validation, verify reports, and ensuring the information on the app

is up-to-date. The Super-Admin has broader access to manage resources, managing donations, and oversee the entire system.

Once the data is validated and processed, it becomes visible in key user-facing modules, ensuring that vital information is accessible and up-to-date. These modules include details about Evacuation Centers, Barangay Contact Information, News Updates, and Maps highlighting hazard-prone areas and available emergency resources. Through this integration, residents can easily locate the nearest evacuation sites, contact local authorities for assistance, and stay informed about ongoing situations such as road closures, weather advisories, or relief operations. By providing real-time and reliable data, the system empowers users to make informed and timely decisions that enhance their safety and preparedness during emergencies. This seamless flow of verified information also promotes transparency, coordination, and quick response among the community and local officials.

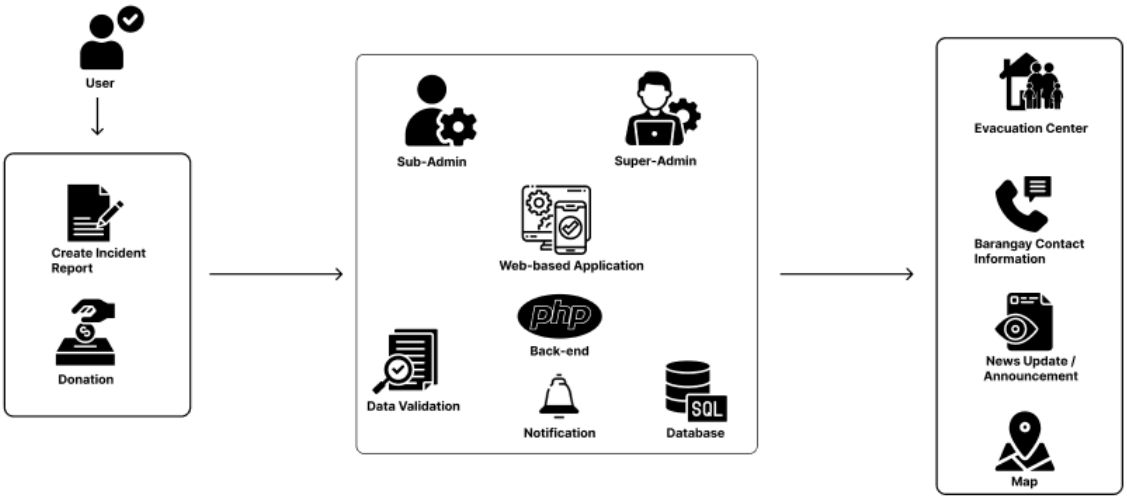


Figure 1. Conceptual Framework

CHAPTER 3

METHODOLOGY

This chapter focuses on project design, methods for developing web-based applications, and the necessary data.

Project Design

The Disaster Ready web-based application is structured into modules accessible by three roles: Super Admin and Sub Admin, all of whom access the system through the Login Module. This allows account creation, password recovery, and login. Users can explore several key features. The Calamity Module educates about disaster types and safety tips. The Emergency Resources Module provides local hotlines and barangay-specific resources. The Updates and News Module updates weather, road conditions, and evacuation center status. Through the Incident Report Module, users can send incident reports to the admin. The Disaster Risk Mapping Module shows evacuation routes and risk zones, while the Prone Areas Module highlights flood- and landslide-prone areas. Users can also donate via various methods in the Donation Module, learn more about the app in the About Us Module, and manage their accounts in the Account Settings Module.

Super Admins have full control. Their Dashboard shows overall data such as users, evacuation centers, and emergency reports. They manage evacuation centers, update maps, track relief goods distribution, and edit barangay contact information. They can post news, manage donation platforms, oversee users, and receive emergency notifications. Sub Admins have similar access but cannot manage users or donation platforms. Lastly,

all modules are connected to a central Database, ensuring all data is synchronized and secure for effective disaster response and management.



Figure 2. System Architecture

Project Locale

This study was conducted in the Municipality of Los Baños, Laguna, a town known for its academic institutions, residential communities, and frequent exposure to natural disasters such as typhoons, floods, and earthquakes. Due to its geographical location and surrounding natural

features, Los Baños is vulnerable to environmental hazards that impact public safety and infrastructure.

3The area was chosen as the project locale to develop "Disaster Ready," a Web-based Application designed to assist residents and local authorities in managing evacuation efforts and locating essential resources during calamities. The system aims to improve real-time communication, accessibility of emergency data, and coordination between barangay officials and the community.

Project Development

Software Development Model

The development of software applications typically follows a structured series of steps, commonly known as the Software Development Life Cycle (SDLC). This cycle is designed to guide developers through the necessary processes to build and deliver functional software solutions effectively. The SDLC typically encompasses key stages, including Planning, Requirements, Design, Development, Testing, Deployment, and Maintenance. These stages are often divided into six to eight steps, depending on the complexity and scale of the project.

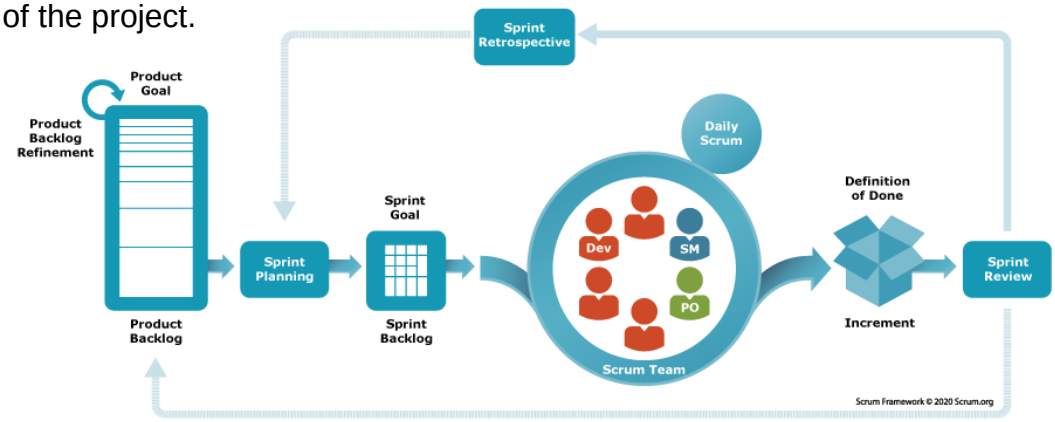


Figure 3. Agile Scrum

Agile Scrum is a widely recognized software development methodology that emphasizes delivering software products through iterative and incremental progress. This approach is highly effective in promoting teamwork, enhancing flexibility, and ensuring customer satisfaction. By adopting Agile Scrum, development teams can adapt to changing requirements while maintaining consistent progress throughout the project lifecycle.

The Agile Scrum framework is designed to improve collaboration among team members and stakeholders. It encourages continuous communication, frequent feedback, and adaptability to changes. This methodology is widely embraced across the software industry because it delivers results efficiently. In this study, the Agile methodology was chosen as the preferred approach to identify the required development steps for the project. The Agile framework ensures that tasks are broken down into manageable segments, which are completed through structured iterations known as sprints**.** Each sprint follows a defined timeframe, allowing the team to focus on specific goals and deliverables.

Key roles in the Agile Scrum framework include the Product Owner, who is responsible for defining the project vision, setting priorities, and determining what the team should develop. They manage the product backlog, which contains a detailed list of features, enhancements, and requirements that need to be addressed, ensuring alignment with customer expectations. The Agile Master (or Scrum Master) acts as a facilitator, guiding the team through the development process by eliminating obstacles that could hinder progress and ensuring the team remains focused and productive. Lastly, the

Development Team is composed of skilled developers who collaborate to build the application. They work closely with the Product Owner and Agile Master to deliver the required features, following the sprint plan and conducting regular meetings to assess progress, address challenges, and maintain alignment with project goals.

The Agile Scrum methodology relies heavily on the concept of sprints**,** which are time-boxed development cycles where specific tasks and features are completed. Each sprint begins with a planning session where the team identifies the objectives for that period. During the sprint, progress is monitored through daily meetings, ensuring any issues are addressed promptly. The product backlog serves as the primary reference for guiding the team's efforts. It outlines the necessary features and improvements, allowing the team to stay organized and focused. By working in short, controlled sprints, the Agile Scrum framework promotes continuous progress and helps teams adapt to changing requirements. This structured yet flexible approach makes Agile Scrum an effective method for ensuring that software development projects are completed efficiently while meeting the desired goals and user expectations.

Product Backlog

The Product Backlog for the Disaster Ready web-based application serves as a comprehensive roadmap that guides the continuous development of the system to enhance disaster preparedness, communication, and response in the Municipality of Los Baños. It prioritizes core modules such as Incident Reporting, Updates and News, and the Evacuation Center, which

directly support residents during emergencies by providing reliable information and quick access to safe routes and assistance. Supporting modules like Donation Management, Calamity Types and Safety Tips, and Barangay Contact Information promote awareness, transparency, and community participation, ensuring that residents are informed and that relief efforts are organized effectively.

Additionally, administrative tools such as User Management, News and Updates, and Relief Goods Distribution strengthen coordination among Super Admins and Sub-Admins, maintaining accurate data management and efficient communication. Technical priorities, including database optimization and performance tuning, ensure smooth functionality, accessibility, and real-time responsiveness even during low-connectivity situations. The backlog remains adaptive, evolving through stakeholder feedback, usability testing, and emerging community needs to ensure that Disaster Ready remains a responsive, user-centered, and dependable platform for disaster management and recovery.

Beyond immediate disaster response, the backlog also emphasizes preparedness and resilience-building initiatives that empower residents long before a calamity occurs. Features such as preparedness checklists, scheduled disaster drills, community training modules, and risk assessment tools aim to cultivate a culture of readiness throughout the municipality. These enhancements help bridge the gap between emergency awareness and actionable knowledge, ensuring that individuals and families are equipped with the skills needed to respond effectively when disasters strike.

Table 2**.** Super Admin Product Backlog

Priority	Feature	Story
1	Login / Registration	I want to create an account and log in
2	Dashboard	I want to view the overview of each evacuation center in the municipality of Los Baños.
3	Evacuation Center	I want to create and update the evacuation centers' status.
4	Disaster Risk Mapping	I want to update the road status in case of flooding or fallen trees.
5	Relief Goods	I want to know the list of families per barangay to facilitate the distribution of relief goods.
6	Barangay Contact	As an Admin, I want to update the latest contact information of each barangay.
7	Updates and News	As a Super Admin, I want to publish news to keep the residents updated on what's happening in the community.
8	Donation	As a Super Admin, I want to view the history of donations form the donors.
9	User Management	I want to view the list of the registered accounts of Sub admins and Residents.
10	Account Setting	I want to update the information of my account in case there are changes to the data.

Table 3**.** Sub Admin Product Backlog

Priority	Feature	Story
1	Login / Registration	I want to request an account and log in
2	Dashboard	I want to view the overview of each evacuation center in the municipality of Los Baños.
3	Evacuation Center	I want to create and update the evacuation centers' status.
4	Disaster Risk Mapping	I want to update the road status in case of flooding or fallen trees.
5	Barangay Contact	As an Admin, I want to update the latest contact information of my barangay.
6	Updates and News	As a Super Admin, I want to publish news to keep the residents updated on what's happening in the community.
7	User Management	I want to view the list of the registered accounts of Sub admins.
8	Account Setting	I want to update the information on my account in case there are changes to the data.

Table 4**.** Resident Product Backlog

Priority	Feature	Story
1	Homepage	I want to view a clear and visually appealing homepage with key information about the app's purpose and features.
2	Calamities	I want to view types of calamities and calamity tips to follow before, during and after the disaster.
3	Contact Information	I want to view the information per barangay and public hotlines.
4	Updates and News	I want to view the updates and news about the community
5	Prone Areas	I want to know which areas of Los Baños are prone to flooding and landslides, and what precautions should be taken when in those areas.
7	Donate Now	I want to see the available donation options to donate.
8	Report Incident	I want to send a report to the admin if there is an incident in our barangay or wherever I may be around the municipality of Los Baños.

Sprint Planning

The Municipality of Los Baños is a densely populated area that is vulnerable to a range of natural disasters, including typhoons, floods, landslides, volcanic eruptions, and earthquakes. To address this, the "Disaster Ready" web-based application was developed to assist residents and local authorities in managing evacuation procedures and locating essential resources during calamities. This application would feature an interactive map that shows the nearest evacuation centers based on the user's real-time location. The map would display icons indicating safe routes, evacuation points, and resource hubs to facilitate efficient, secure navigation during emergencies.

Additionally, the application would include an "Incident Report" button that allows users to submit incident reports directly to the admin. Once notified, the admin would track the location of the user’s emergency report. It reviews and validates the report before posting it as news to inform other users about the situation. This feature aims to enhance communication and ensure timely updates, thereby facilitating better disaster response and community safety. The development team would prioritize these critical features during the sprint to ensure functionality, accuracy, and user accessibility.

Sprint Backlogs

The estimated plans and designs of the user interface that would be made to create the web-based disaster risk reduction application are outlined below. Each sprint backlog details the corresponding tasks, ensuring organized development and proper time allocation to meet project goals efficiently.

Table 5. Super Admin and Sub Admin Sprint Backlogs (Development)

Priority	Activity Sprint	Task	Total Hours per Day
1	I want to create an account and log in	Meet with the teammates to brainstorm ideas for the Log in Module. Design the Login Module. Code the Login Module	3hrs
2	I want to send Updates and news to the Residents.	Meet with the teammates to brainstorm ideas for the Updates and news Module. Design the Updates and news Module. Code the Updates and news Module.	4hrs
3	I want to create and update the evacuation centers' status.	Meet with the teammates to brainstorm for the Evacuation Center feature. Design the Locate Evacuation Center Module. Code for the Locate Evacuation Center Module.	6hrs
4	I want to update the latest contact information of my barangay.	Meet with the teammates to brainstorm the Barangay Contact Information Module. Design Barangay Contact Information Module. Code the Design Barangay Contact Information Module.	4hrs
5	I want to update the road status in case of flooding or fallen trees.	Meet with the teammates to brainstorm the Disaster Risk Mapping Feature. Design the Disaster Risk Mapping Feature. Code the Disaster Risk Mapping Feature.	6hrs
6	I want to update the information on my account in case there are changes to the data.	Meet with the teammates to brainstorm the Account Setting Feature. Design the Account Setting Feature. Code the Account Setting Feature.	3hrs

Table 6. Resident Sprint Backlogs (Development)

Priority	Activity Sprint	Task	Total Hours per Day
1	I want to view the Updates and News about the community	Meet with the teammates to brainstorm ideas for the Updates and News Module. Design the Updates and News Module. Code the Updates and News Module	7hrs
2	I want to send a report to the admin if there is an incident in our barangay or wherever I may be around the municipality of Los Baños.	Meet with the teammates to brainstorm for the Incident Report Module. Design the Incident Report. Code for the Incident Report Module.	5hrs
3	I want to view the status of evacuation centers, including the capacity, vacancy, and occupancy.	Meet with the teammates to brainstorm the Evacuation Center Module. Design Evacuation Center Module. Code the Evacuation Center Module	6hrs
4	I want to view the emergency resources per barangay.	Meet with the teammates to brainstorm the Barangay Contact Information Module. Design Barangay Contact Information Module. Code the Barangay Contact Information Module.	4hrs

Sprint Execution

The sprint execution begins once the development team has finalized the Sprint Backlog for the "Disaster Ready" web-based application. At this stage, the team focuses on implementing key features such as user authentication, evacuation center mapping, and real-time resource tracking. The sprint is carried out with minimal interruptions, allowing the team to concentrate on completing the planned tasks efficiently. However, any urgent

modifications or unexpected challenges that arise are noted and scheduled for the next sprint cycle.

After completing each sprint, the team presents the implemented features to stakeholders, including local government units and emergency responders, to gather feedback. This feedback is then analyzed to determine necessary improvements or refinements. The team ensures that critical adjustments are prioritized in the following sprint while maintaining the overall project timeline. This iterative approach guarantees that the application remains functional, user-friendly, and aligned with the needs of the Municipality of Los Baños in times of natural calamities.

Flowchart

The flowchart provides a comprehensive overview of the user journey and system flow within the Emergency and Disaster Management System, outlining interactions for three main user types: Regular Users, Super Admins, and Sub Admins. The process begins at the Start point, directing all users to the Homepage, which offers access to general features such as Calamities, Evacuation Center, Disaster Risk Mapping, Contact Information, Prone Areas, Updates and News, and an About Us section. If the user is identified as a Regular User, they are taken to a user-friendly interface that provides access to modules such as Calamities, Evacuation Center, Disaster Risk Mapping, Contact Information, Prone Areas, Updates and News, Incident Report, and About Us. These modules primarily serve to inform users about disasters, provide emergency contact resources, and allow them to report incidents.

If the user logs in as a Super Admin, they are directed to the Admin Dashboard, which grants full control over the platform's backend functions. Super Admins can manage critical modules, including Evacuation Centers, Relief Goods, Disaster Risk Mapping, User Management, Updates and News, Barangay Contact Information, Donations, Incident Report, and Account Settings. Each of these modules includes specific sub-functions such as adding evacuation centers, viewing distribution history, managing announcements, updating contact details, and overseeing both user accounts and system activities.

For Sub Admins, a slightly limited but still administrative role is provided. Upon login and role identification, Sub Admins are directed to a dedicated dashboard that includes access to core modules similar to those of the Super Admin but with fewer privileges. They can manage modules such as Evacuation Centers, Relief Goods, Disaster Risk Mapping, Barangay Contacts Information, Updates and News, Notifications, and Account Settings. However, advanced administrative tasks like User Management and Donation Management may be restricted or read-only, depending on the system's permission levels. Like Super Admins, Sub Admins are responsible for keeping the system updated with real-time data and relevant announcements, ensuring smooth coordination during emergencies.

Both Super Admins and Sub Admins have sub-functions for each module (as shown in the flowchart), allowing them to perform tasks such as adding data, viewing reports, and managing emergency responses. The system processes for each user type conclude at their respective End nodes, indicating the completion of their interaction with the system. This flowchart

effectively distinguishes between user roles, illustrating a structured and role-based navigation system designed to streamline disaster response operations and ensure information accuracy across all access levels.

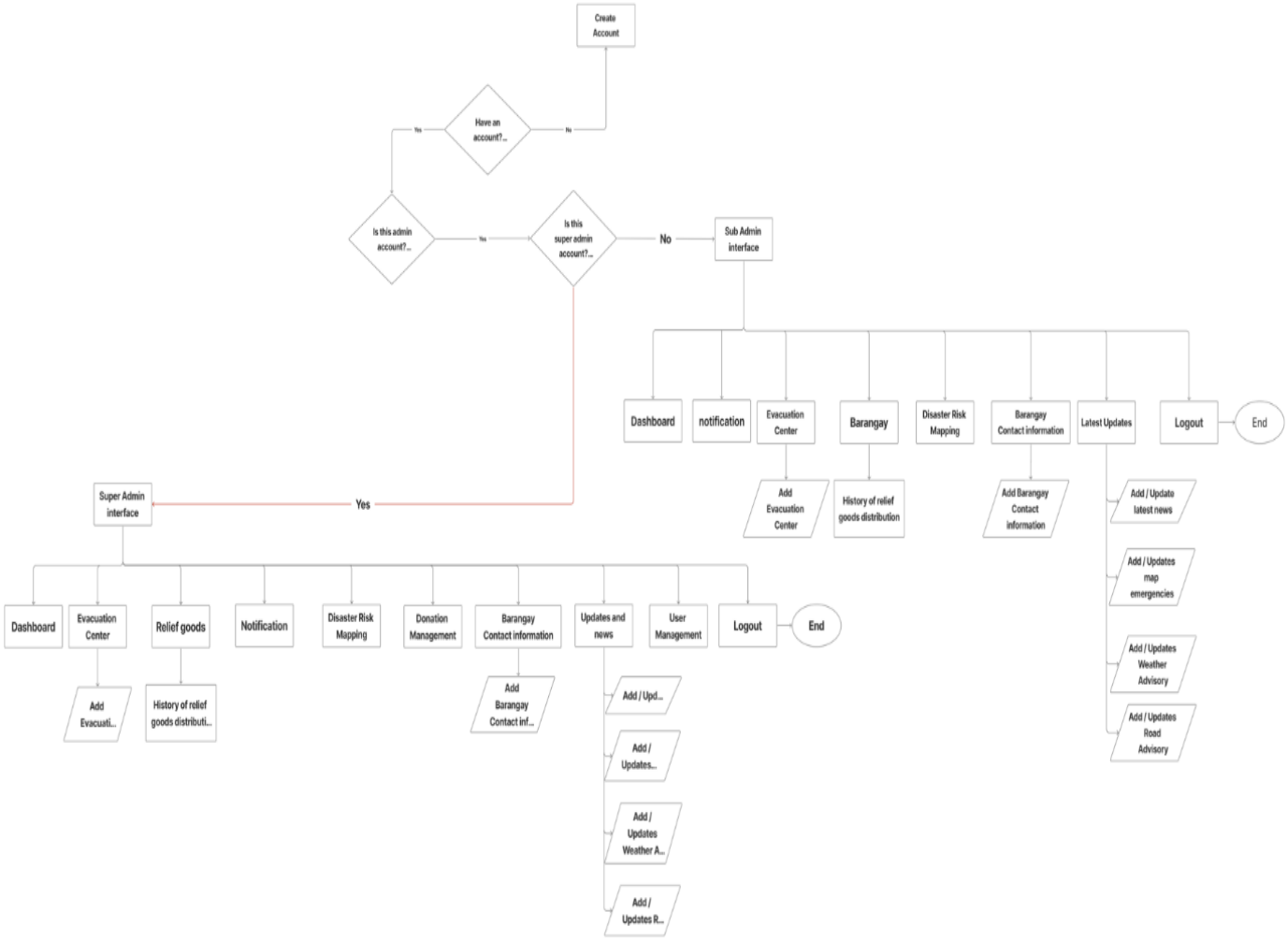


Figure 4. System Flowchart

Use Case Diagram

The image below presents a use case diagram for a disaster management system designed to assist different user roles: Master Admin, Sub Admin (Barangay). The Master Admin has the most comprehensive access, including system overview, managing evacuation center details, distributing relief goods, updating maps and emergency contact details, managing

donations, posting news and guidelines, handling user accounts, and configuring the system. The Sub Admin shares similar functions but is limited to their barangay scope, allowing them to manage local evacuation centers, update barangay-specific information, and provide news or updates relevant to their area.

For regular Users, the system offers features focused on information and reporting. They view the homepage, learn about different calamities and safety tips, see emergency resources, view details about the system, donate, access prone areas, and view updates and news. Users can also report incidents or emergencies. The diagram highlights how each role interacts with the system, ensuring that both administrative users and residents have the tools necessary for effective disaster response and preparedness.

In addition to defining user access, the use case diagram also emphasizes the flow of information from the community to administrators, supporting faster and more accurate decision-making during critical situations. By allowing residents to directly report emergencies, the system ensures that barangay-level admins can immediately respond, verify incidents, and communicate these reports to the Master Admin when necessary. This structured hierarchy of information flow helps prevent delays and ensures that emergency teams are dispatched efficiently based on verified and up-to-date information.

Moreover, the diagram illustrates how the system serves as a centralized platform for disaster-related communication, reducing fragmentation and confusion during calamities. By consolidating evacuation data, safety

guidelines, barangay announcements, and relief distribution updates in one platform, users are guaranteed access to consistent and reliable information. This strengthens coordination not only between government units but also within the community, ultimately contributing to a more organized, informed, and resilient disaster management ecosystem for the Municipality of Los Baños.

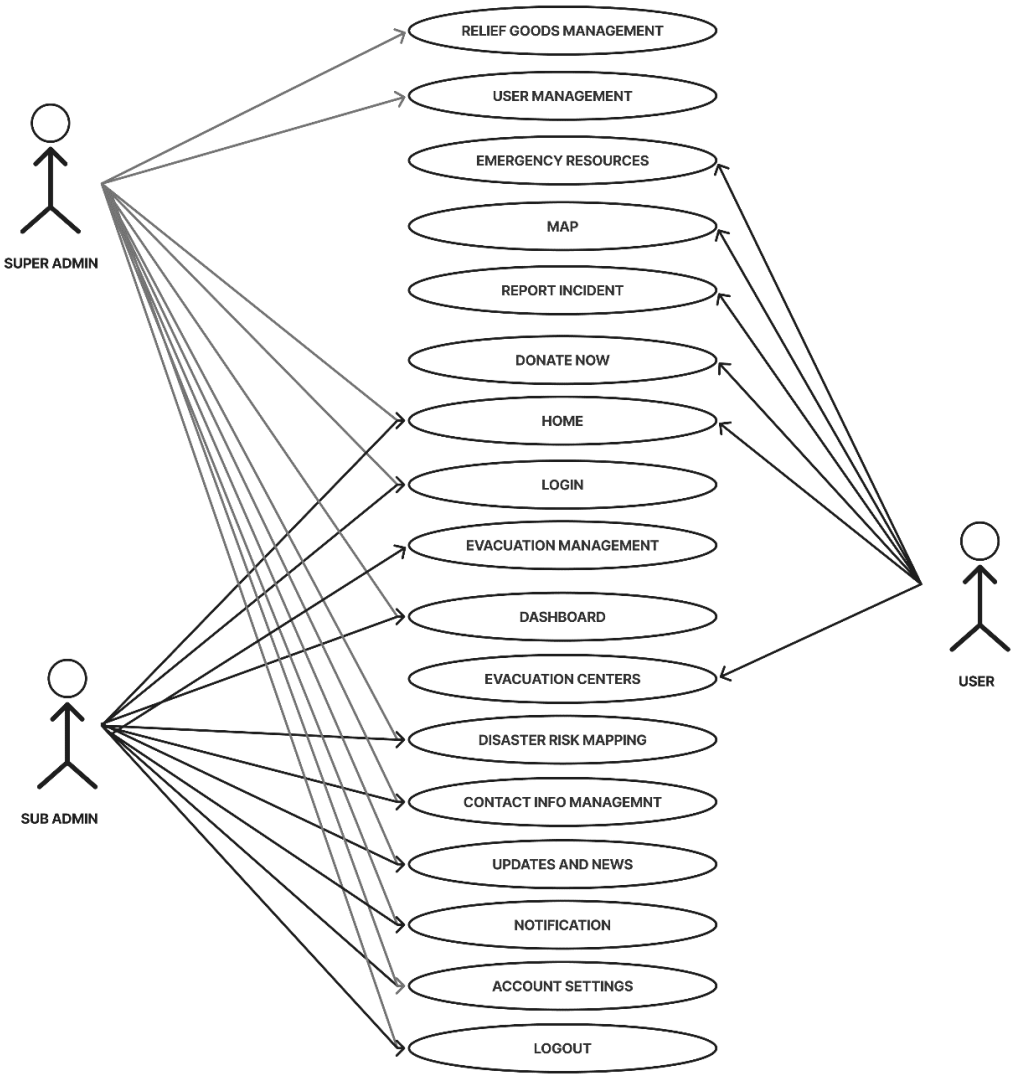


Figure 5. Use Case Diagram

Geographic Information System (GIS)

A Geographic Information System (GIS) is a technology used for gathering, managing, analyzing, and visualizing spatial or geographical data. It enables the mapping and interpretation of data based on location, helping users to see patterns, relationships, and trends across geographic space. GIS integrates layers of information about a place to give users a better understanding of that location. These layers may include maps, satellite imagery, coordinates, and tabular data linked to specific geographical points. GIS plays a crucial role in disaster preparedness and management. Using GIS, critical data such as hazard zones, evacuation routes, emergency facilities, and affected areas can be plotted on a digital map in real-time. This allows citizens and government agencies to quickly make informed decisions, which is vital during emergencies such as typhoons, earthquakes, or floods. With Disaster Ready, GIS is a core component of the system. The application utilizes GIS to provide users with interactive maps that display real-time updates on disasters, such as flooded areas, damaged roads, and active evacuation centers. By incorporating Google Maps JavaScript API and Google Places API, the system delivers accurate location-based information that is essential for both citizens seeking help and responders coordinating relief efforts.

Moreover, GIS in Disaster Ready enhances public awareness by allowing users to report incidents by tagging their location. These reports are then visualized on the map for other users to see, creating a community-driven disaster alert system. The ability to analyze geographic patterns, such as the frequency of disasters in specific areas, also aids in long-term planning and

risk reduction for local government units. Ultimately, GIS empowers Disaster Ready to transform static emergency data into interactive, actionable intelligence, fostering a more resilient and informed community.

Materials

To develop the Disaster Ready, various development tools and technologies were utilized. These materials supported both the frontend and backend functions of the system, enabling real-time updates, interactive maps, and efficient data management. Each tool plays a specific role in ensuring that the application is responsive, user-friendly, and reliable during emergency situations. Below are the materials used in building the system and their relevance to the project.

HTML

HTML is used to structure the web application's content, such as announcements, emergency reports, and disaster-related information. It provides the foundation for displaying maps, safety tips, and other critical content to users during calamities.

JavaScript

JavaScript enables interactive features of the system, such as real-time updates, map interactions, and dynamic data displays (e.g., evacuation status, weather alerts). It helps make the app responsive and functional in emergency conditions.

Tailwind CSS

Tailwind is a utility-first CSS framework used to build a clean, mobile-responsive, and fast UI. It helps in designing a visually consistent layout

across devices, which is important for residents accessing the app on phones or tablets during disasters.

PHP

PHP is used as the server-side language to handle backend operations like storing reports, managing users, and processing donations. It ensures secure communication between the frontend and the database, critical for sensitive emergency data.

MySQL

MySQL is the database system used to store and manage data such as user accounts, emergency reports, evacuation centers, and donation history. It plays a vital role in organizing and retrieving information quickly during disaster response.

Leaflet Map

Leaflet is an open-source JavaScript library that enables the creation of interactive maps for web-based applications. In the Disaster Ready project, it is used to display evacuation centers, affected barangays, and nearby resources in Los Baños, helping users easily locate safe areas and access real-time geographic information during disasters.

Easysend SMS

It is an API-based messaging platform used to send real-time alerts and notifications. In the Disaster Ready, it enables the system to send SMS messages to residents who have submitted an incident report, ensuring they

receive important updates even without internet access once the admin has received and taken action on their reported concern.

PHP Mailer

is a PHP library designed to simplify the process of sending emails directly from a web-application. In this project, it is utilized to send verification emails, and system notifications**,** improving communication between users and administrators.

Vlucas/phpdotenv

is a PHP library that manages environment variables stored in a.env file. It is used in the project to securely handle sensitive information such as API keys, database credentials, and configuration data, enhancing overall security and maintainability.

Postman

is a software tool for testing and managing APIs. The developers used it in the Disaster Ready project to test API endpoints, validate responses, and ensure system reliability during the integration of features like SMS alerts and authentication.

GitHub

is a cloud-based version control and collaboration platform for developers. It was used in this project to store source code, track changes,

and collaborate efficiently, ensuring organized and continuous development throughout the system's creation.

Hardware

The table outlines the minimum hardware specifications required to run the Disaster Ready web-based application efficiently. A system with 8 GB of RAM ensures smooth multitasking and performance during real-time data processing. A 256 GB SSD offers fast storage access, which is essential for handling user reports, map data, and emergency updates. Lastly, the application is optimized to run on Windows 11, ensuring compatibility with modern system features and security updates.

Table 7. Hardware Requirements

Component	Specification
Random Access Memory (RAM)	4 GB
Storage	256 GB SSD
Operating System	Windows 9

Table 8 highlights key components and their necessary specifications. For example, the operating system must be Android 10/iOS 6 or higher, a minimum of 3 GB of RAM is needed, and the processor should be a quad-core 1.8 GHz or faster. Sufficient storage (at least 64 GB with 500 MB free) and a display of at least 5.5 inches with 720p resolution are also requirements. Finally, 4G LTE and Wi-Fi connectivity, along with built-in GPS, are essential for functionality. These specifications ensure the mobile device can handle the demands of the application without performance issues.

Table 8. Minimum Requirements for Mobile Phone

Component	Specification
Operating System	Android 10 / iOS 6 or higher
RAM	3 GB
Processor	Quad–core 1.8GHz or higher
Storage	At least 64 GB (with 500MB free)
Display minimum	5.5 inches, 720p resolution
Connectivity	4G LTE, Wi-Fi enabled
GPS	Built-in GPS

Project Testing and Evaluation

Software Testing

Software testing emphasizes its systematic, multi-level approach. It highlights that software testing is designed with different levels to ensure thorough checking of behavior and performance throughout the system development life cycle (SDLC). A diagram illustrates the process, showing the flow from unit testing to integration, system, and acceptance testing. The figure also shows that in a specific system, researchers would test modules, ensuring alignment with intended functionalities. Each phase of the SDLC undergoes testing to identify any missing areas between development cycles. The levels of software testing mentioned are unit, integration, system, and acceptance testing. A diagram illustrates the process, showing the flow from unit testing to integration, system, and acceptance testing. The figure also shows that in a specific system, researchers would test modules, ensuring alignment with intended functionalities.

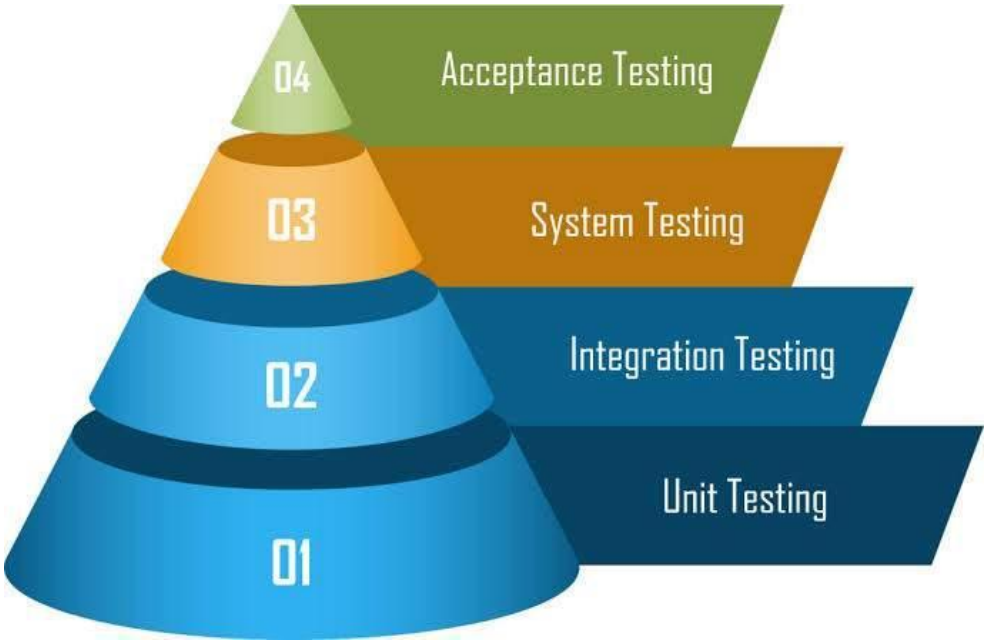


Figure 6. Levels of Software Testing

Unit Testing

Figure 8 illustrates the Unit Test Life Cycle, a process that plays a critical role in ensuring the reliability and quality of software development, particularly relevant to our capstone project on disaster management. The cycle begins with checking out code from the repository, where developers retrieve the latest version of the system. Next, they make changes to enhance features or fix issues, such as updating evacuation center details or improving the donation tracking system. Once changes are made, unit tests are executed to verify that individual parts of the code function correctly.

If any defects are identified, the developer would fix the issues and re-execute the unit tests to ensure the problem is resolved. After passing all tests, the code undergoes a code review, where team members verify its quality and logic. Finally, the verified code is checked back into the repository, completing

the cycle. Applying this cycle to our capstone project ensures that each module —such as real-time updates, report generation, or user management —is reliable and error-free, which is vital in a system supporting emergency preparedness and response.



Figure 7. Unit Testing

Integration Testing

Figure 8 illustrates the concept of Integration Testing, where two or more individual modules—Module A and Module B—are tested together to ensure they work correctly as a combined unit. In the context of our capstone project on disaster management, integration testing is crucial because our system consists of interconnected modules such as User Management, Real-time

Updates, Emergency Report Tracking, Barangay Contact Information, and Evacuation Center Monitoring.

For example, when the Admin updates evacuation center information (Module A), it should correctly reflect in the real-time updates module (Module B) that is visible to residents. Integration testing ensures that this interaction happens smoothly and without errors. By thoroughly testing how modules communicate and work together, we prevent system failures and ensure that critical features, especially those used during emergencies, function reliably across the entire platform.

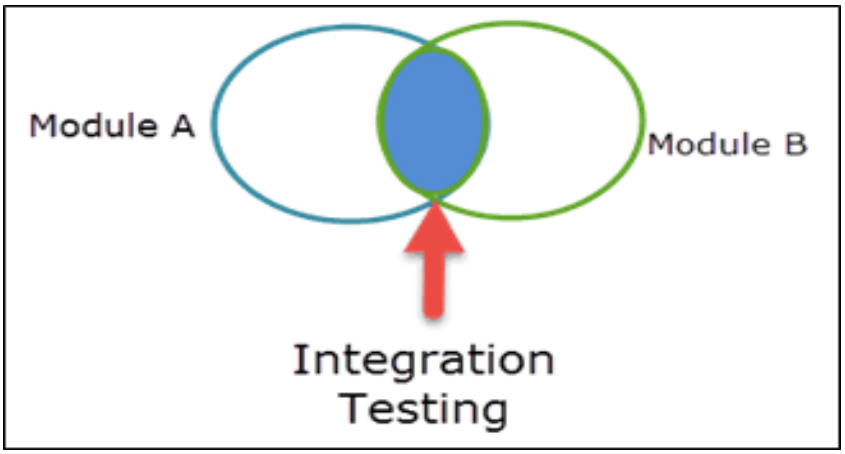


Figure 8. Integration Testing

API Testing

Figure 9 demonstrates how an API (Application Programming Interface) facilitates communication between the Application/Client, the Server/Data Source, and back. In the context of our disaster management capstone project, APIs are crucial for real-time functionality. For instance, when a resident submits an emergency report, the application sends an API request

to the server, which processes the data and stores it. Through the API, the admin can track who sent the emergency report, view relevant details, and respond promptly.

Additionally, APIs are used to fetch and display the location of evacuation centers on a map for users. This allows residents to find safe locations nearby during a disaster easily. The real-time interaction enabled by APIs ensures that information flows quickly and accurately between users and administrators, improving both response time and overall disaster preparedness.

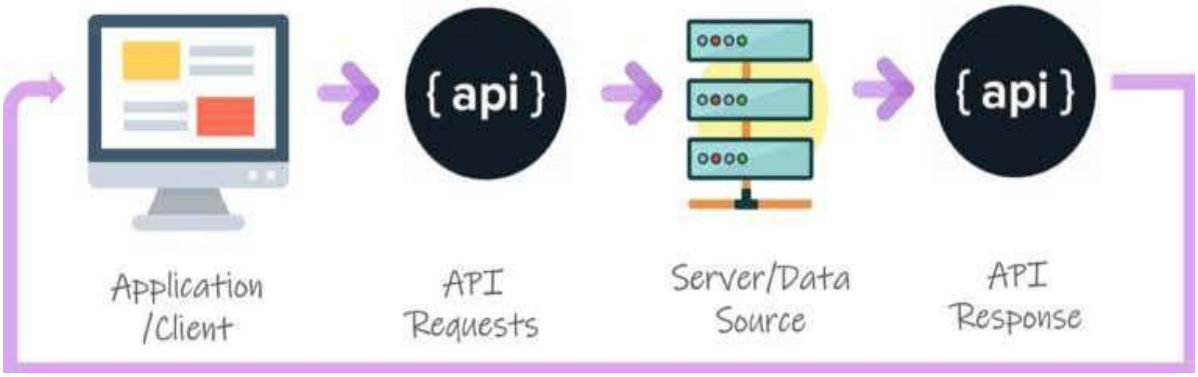


Figure 9. API testing

System Testing

System testing for Disaster Ready involves thoroughly evaluating all features and modules to ensure they function correctly under real-world conditions. This includes testing the account registration, emergency reporting, donation platform, real-time updates, disaster risk maps, and resource locator tools across both web and mobile versions. Special attention

is given to the responsiveness and accuracy of the Google Maps API integration for displaying evacuation centers and prone areas. Admin functionalities like data validation, user management, and news announcements are also tested for proper access control and reliability. The goal of system testing is to identify and fix bugs, ensure a smooth user experience, and validate that the system supports timely and accurate communication during emergencies.

Compatibility Testing

The compatibility testing for the Disaster Ready web-based application would be conducted to ensure that the system operates effectively across different devices, browsers, and operating systems. This testing aims to verify that all modules, such as the Evacuation Center Locator, Real-time Updates, Donation Management, and Incident Reporting, and etc. function consistently and display correctly regardless of the platform used. The application would be tested on commonly used browsers like Google Chrome, Mozilla Firefox, and Microsoft Edge, as well as on both desktop and mobile devices, to assess responsiveness and interface adaptability.

Testing and Evaluation

The image contains a diagram showing the ISO 25010 model, categorizing software product quality into several characteristics, including Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, Security, Maintainability, and Portability. Each characteristic is further broken down into sub-characteristics. For evaluating the effectiveness of Disaster Ready, the researchers would do a beta test with a total of one

hundred (100) residents of Los Baños, including the admins, and two (2) IT specialists. The ISO 25010 quality model was used as a standard by the researchers for the Likert scale. The quality model determines which quality characteristics would be taken into account when evaluating the properties of a software product. Functional suitability is the degree to which the software meets its functional requirements. Efficiency, which characterizes the software performance in its intended function in a timely and resource-efficient manner. Compatibility, where the software can operate with other systems or components. Usability, a quality characteristic that rates the software as easy to use and understand. Reliability, when the software performs without failures or errors. Security, the protection against unauthorized access and malicious attacks. Maintainability, the software could be modified or adapted. And lastly, Portability, where the software can be transferred from one environment to another. Likert Scalar was answerable by four responses, which are 4. = Strongly Agree, 3 = Agree, 2 = Disagree, and 1 = Strongly Disagree. The ISO 25010 model sax useful for evaluating software quality because it provides a comprehensive framework that covers all aspects of software quality. By evaluating the software based on the ISO 25010 model, organizations could identify areas for improvement and make data-driven decisions about how to allocate resources to improve software quality.



Figure 10. ISO 25010 Model

Population of the Study

Based on the study's findings, the researchers conclude that the Disaster Ready web-based application is an effective and reliable platform for enhancing disaster preparedness and response in the Municipality of Los Baños. By centralizing communication among the municipal government, barangay sub-administrators, and residents, the system enables timely and organized dissemination of critical information during emergencies. The integrated navigation and incident reporting features support safer decision-making by helping users identify secure routes and locations, thereby strengthening coordinated disaster response. Overall, the implementation of Disaster Ready has important implications for local disaster management, as it supports evidence-based planning, improves community awareness, and contributes to greater resilience and preparedness against natural calamities.

Sampling Design

For this study, purposive sampling was employed to select participants most relevant to the objectives of the Disaster Ready application. The respondents were divided into two groups: (1) end-users, consisting of residents of Los Baños who are most likely to be affected by natural calamities such as typhoons, floods, and earthquakes, and (2) expert evaluators, including barangay officials and emergency responders who are directly involved in disaster preparedness and management. The end-users evaluated features such as the evacuation center locator, navigation routes, and emergency reporting for usability and accessibility, while the expert evaluators assessed system functionality, reliability, and overall relevance to disaster response operations. The number of respondents was deemed sufficient

because it provided a diverse representation of both end-users and expert evaluators, ensuring comprehensive feedback on usability, functionality, and relevance. This sample size allowed the researchers to gather meaningful insights while remaining manageable for detailed analysis.

Data Collection Instrument

To gather relevant data for the development and evaluation of the “Disaster Ready” web-based app, the researchers utilized a survey questionnaire as the primary data collection instrument. The questionnaire was designed to assess the current disaster response practices of residents and local authorities, their access to technology, and their specific needs during calamities. It also gathered feedback on system usability and key features after prototype testing. The survey included a combination of multiple-choice, Likert-scale, and open-ended questions, allowing both quantitative and qualitative data to be collected. This instrument was distributed to target respondents such as barangay officials, municipal staff, and residents who are directly affected by natural disasters. The data gathered played a vital role in understanding user expectations and validating the effectiveness of the proposed system in enhancing disaster preparedness and response in the Municipality of Los Baños.

Statistical Treatment

The data collected through the survey questionnaires were carefully organized, tabulated, and statistically analyzed to assess the overall effectiveness, performance, and quality of the “Disaster Ready” web-based application. The evaluation focused on selected quality attributes derived from

the ISO/IEC 25010 software quality model, which includes Functional Suitability, Performance Efficiency, Compatibility, Usability, Reliability, and Security. Descriptive statistics, specifically the mean were utilized to determine the level of user satisfaction and agreement regarding the system's functionality, ease of use, and reliability during disaster-related scenarios.

The computed mean scores provided insight into how well the system met user expectations. These statistical tools helped identify the strengths of the application and areas that may require improvement. The results of this analysis served as the basis for concluding the overall quality and effectiveness of the Disaster Ready system, ensuring that it aligns with the objectives of promoting disaster preparedness, timely response, and efficient communication among residents and local authorities.

Additionally, the data analysis process enabled the researchers to determine whether user experiences varied across different demographic groups, such as age, location, or frequency of system usage. This deeper evaluation helped reveal patterns that influence how residents interact with the platform, allowing designers and developers to implement enhancements tailored to the specific needs of various user groups. These findings support data-driven decision-making and ensure that future system improvements address usability gaps while promoting equitable access to disaster-related information.

CHAPTER 4

RESULTS AND DISCUSSION

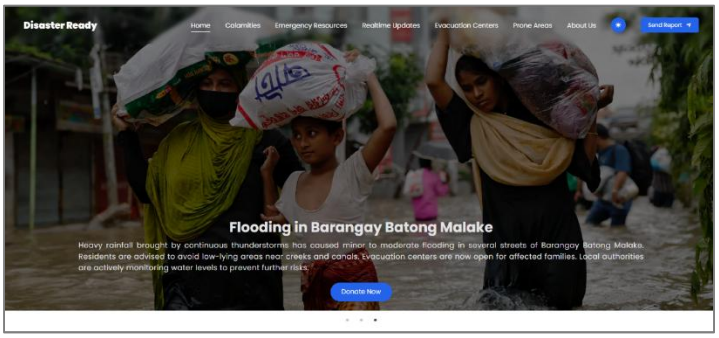
This chapter presents and discusses the findings of the study in relation to the project's objectives. The results were obtained from the survey and system testing conducted to evaluate the Disaster Ready: A Web-based Application for Evacuation and Resource Locator During Natural Calamities in the Municipality of Los Baños. The discussion focuses on the system's usability, functionality, and overall effectiveness based on user feedback and beta testing outcomes.

Results by Objectives

Web-based Application to inform residents about disaster preparedness and partner with local officials for disaster information.

Figures 11(a) and 11(b) show the Home Page of the Disaster Ready web application, which serves as the central hub for accessing vital information about ongoing and recent events in the Municipality of Los Baños. Through timely updates and visual representations of key areas, the Home Page enhances situational awareness and preparedness, enabling users to make faster and more informed decisions during natural disasters. It also features user-friendly navigation that allows easy access to modules such as evacuation center locators, emergency contacts, and safety guidelines, confirming that the system supports efficient interaction even in high-stress situations. The responsive design ensures accessibility across mobile and desktop devices, indicating that the application maintains usability regardless of the platform used. Overall, this module strengthens communication between local authorities and the community by providing a reliable and accessible

information gateway that supports coordinated disaster response and community readiness.



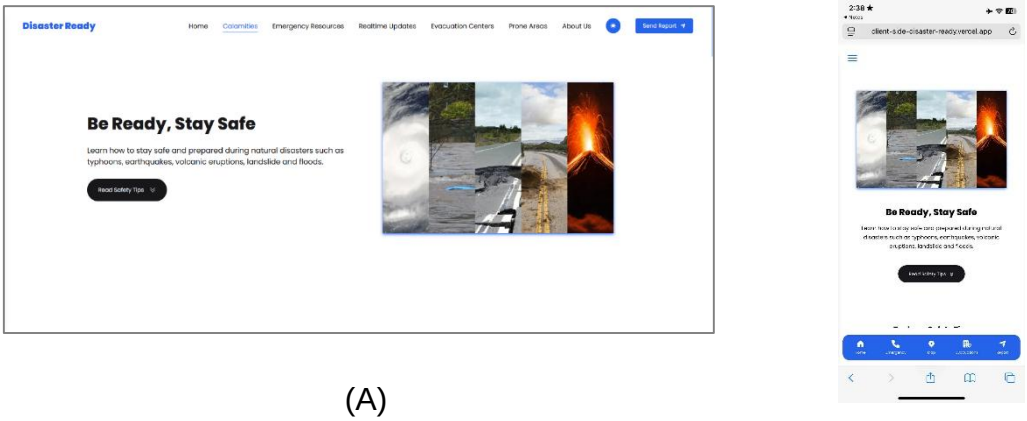
(A)



(B)

Figure 11. Home page

Figure 12 shows the section of the system that provides comprehensive calamity safety tips and practical guidelines to follow before, during, and after natural disasters such as typhoons, floods, landslides, earthquakes, and volcanic eruptions. These tips aim to help residents stay informed, prepared, and alert in any emergency. By offering step-by-step instructions and precautionary measures, the system promotes community awareness and readiness, ensuring that individuals and families know how to respond properly to reduce risks, protect lives, and recover safely after a disaster. This feature also encourages users to continuously update their knowledge on disaster preparedness. It serves as a reliable reference that residents can revisit anytime to reinforce safe practices and improve their overall resilience. This result demonstrates that providing structured, accessible disaster guidance enhances user confidence and encourages proactive safety measures within the community.

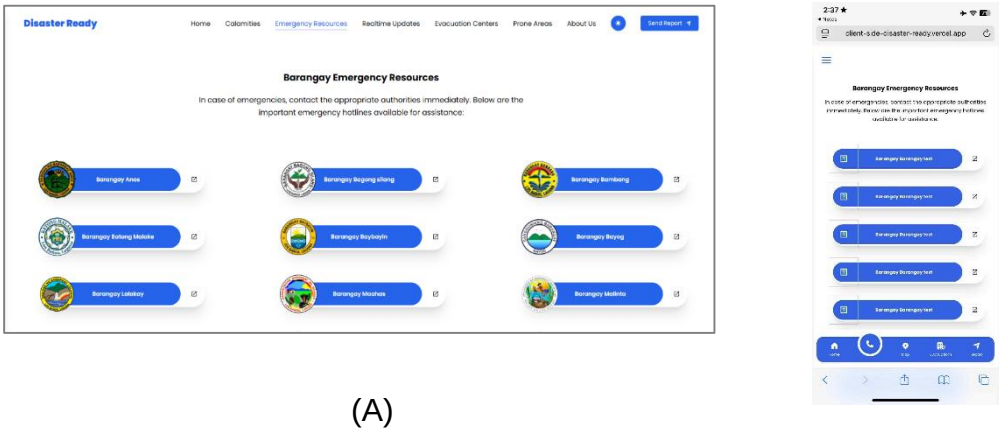


(A)

(B)

Figure 12. Calamity Page

Figure 13 shows the Barangay Emergency Resources module of the Disaster Ready web-based application. It displays a list of barangays in Los Baños with their official logos and quick-access buttons for viewing contact details. When selected, a pop-up displays the barangay's key officials, contact numbers, email address, and Facebook page. This feature helps residents easily reach their barangay during emergencies, improving communication, coordination, and response efficiency. This result demonstrates that centralizing barangay contact information in an accessible interface enhances timely communication and strengthens community emergency response.

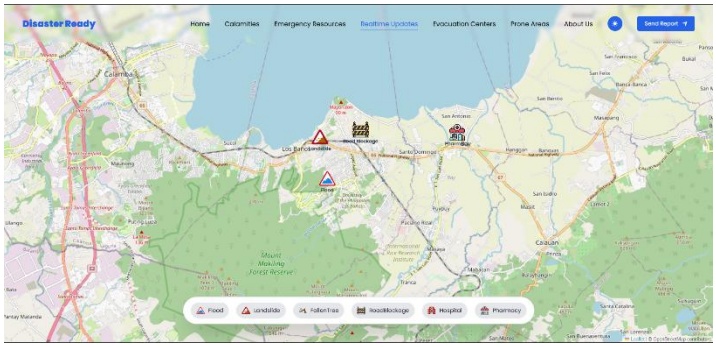


(A)

(B)

Figure 13. Emergency Resources

Figure 14 shows the Real-Time Updates Module of Disaster Ready, displaying ongoing incidents and essential facilities across Los Baños using distinct map markers. This feature enables residents and authorities to quickly assess affected areas and resources, enhancing situational awareness, decision-making, and emergency response. This result demonstrates that real-time spatial visualization improves coordination and accelerates response during disasters.



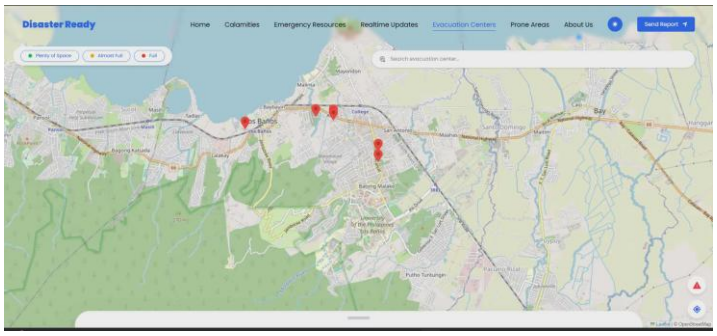
(A)



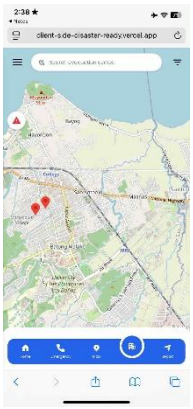
(B)

Figure 14. Map Module

Figure 15 shows the Evacuation Center Module, displaying evacuation sites across Los Baños with color-coded markers indicating real-time occupancy. Residents can quickly locate the nearest safe shelter using the search function, ensuring faster and safer evacuation during emergencies. Local authorities can monitor shelter capacity, manage evacuees, and allocate resources efficiently, improving overall disaster response coordination**.** This result demonstrates that real-time occupancy visualization enhances evacuation efficiency, optimizes resource management, and reduces risks during disaster situations.



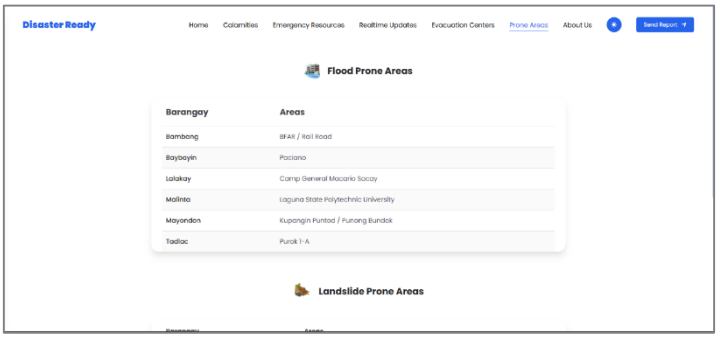
(A)



(B)

Figure 15. Evacuation Center

Figure 16 shows the Prone Areas module, highlighting barangays in Los Baños vulnerable to floods and landslides. Residents and authorities can use this information to plan safer evacuation routes and take preventive measures. This result demonstrates that risk mapping improves risk awareness and strengthens community preparedness**.**



(A)



(B)

Figure 16. Prone Areas

Figure 17 shows the About Us page of the Disaster Ready application, providing background and explaining the purpose behind its development. The page highlights that the app was created in response to recurring natural disasters in the Municipality of Los Baños, such as typhoons, floods, landslides,

earthquakes, and volcanic eruptions. This result demonstrates that clearly presenting the app's purpose and context increases user understanding and engagement, encouraging proactive use of disaster preparedness tools.

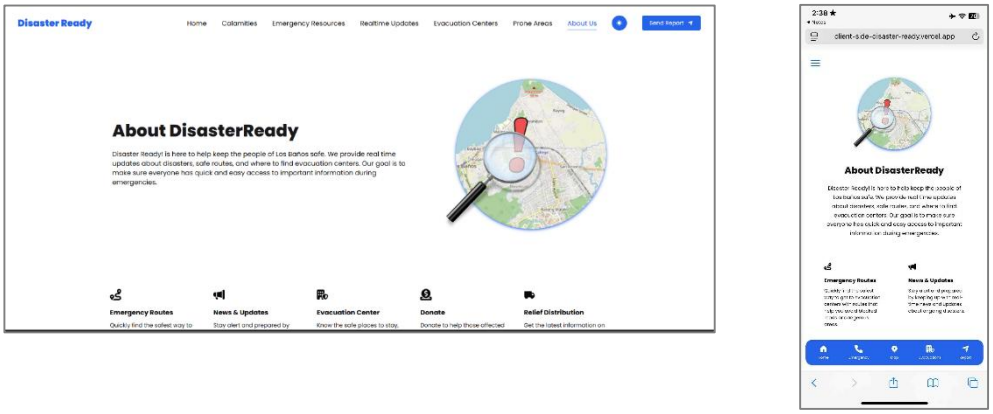


Figure 17. About Us

Figure 18 shows the Send a Report feature, which allows residents to submit incident details and photos during emergencies. Residents can quickly provide critical information about the severity and location of incidents, ensuring authorities receive timely and accurate data. This feature also helps prioritize response efforts and allocate resources more effectively. This result demonstrates that real-time reporting enhances response efficiency and provides authorities with accurate situational information.



Figure 18. Send Report

Real-time information for available evacuation centers and alternative routes using GIS technology, especially during live system operations.

Figure 19 corresponds to Objective No. 2, which aims to provide real-time information for available evacuation centers and alternative routes using GIS technology, especially during live system operations. By integrating Leaflet GIS mapping, the system visualizes the locations and statuses of evacuation centers across Los Baños, allowing users to see which sites have plenty of space, are almost full, or are already at capacity. This real-time visualization helps residents make quick, informed decisions during emergencies, reducing congestion in overcrowded areas. Additionally, the interactive map enhances situational awareness for both users and responders by displaying updated routes and accessibility conditions. The feature also supports authorities in planning and coordinating evacuation strategies more efficiently, ensuring that resources and personnel are allocated where they are needed most. This result demonstrates that the integration of real-time GIS routing significantly reduces uncertainty during evacuation by providing capacity-aware navigation and supporting informed decision-making.

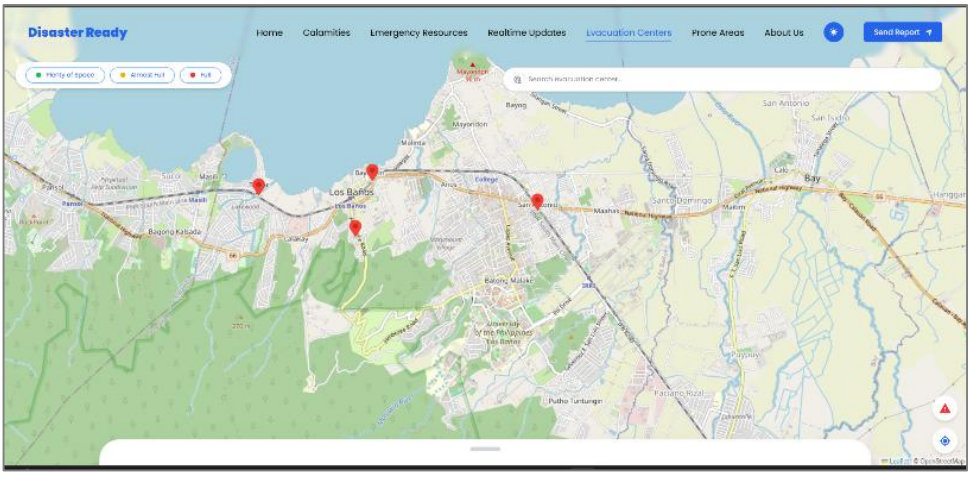


Figure 19. Available Evacuation Center

Figure 20 illustrates the real-time evacuation route feature of the Disaster Ready web-based application. Using GIS and Leaflet integration, the system automatically generates the most efficient route from the user's current location to the selected evacuation center. The map provides detailed step-by-step directions, including distance and turning points, allowing residents to navigate safely during emergencies. This feature not only helps residents avoid congested or hazardous areas but also supports authorities in monitoring evacuation flow and managing traffic during crises. By providing real-time guidance, it increases overall safety, reduces delays, and enhances coordination between the community and emergency responders. This result demonstrates that the integration of real-time GIS routing significantly reduces uncertainty during evacuation by guiding residents along optimized and safe paths.

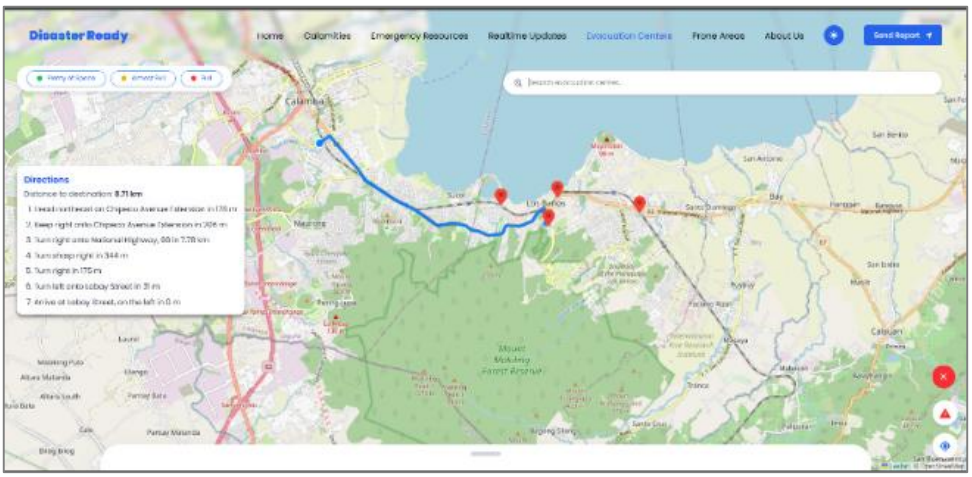


Figure 20. Evacuation Route

Figure 21 illustrates the Evacuation Center List feature of the Disaster Ready web-based application, which provides users with real-time information on available and nearest evacuation centers within Los Baños, including their

locations, current occupancy levels, and accessibility status. Each center is categorized by color-coded capacity indicators—green for “plenty of space,” yellow for “almost full,” and red for “full”—to help residents make informed decisions when seeking refuge. Users can also filter results by barangay or availability status and view detailed information or directions to their chosen center. This feature enhances situational awareness, supports efficient crowd management, and contributes to safer, more organized evacuation processes during emergencies. Moreover, the real-time updates ensure that residents always receive accurate information as conditions change, reducing the risk of overcrowding and helping distribute evacuees more evenly. It also supports local authorities by identifying which centers need additional resources, ultimately enabling faster evacuation times and minimizing potential risks during critical situations. This result demonstrates that real-time monitoring and color-coded capacity indicators enhance evacuation efficiency, reduce congestion, and support informed decision-making for both residents and emergency responders**.**

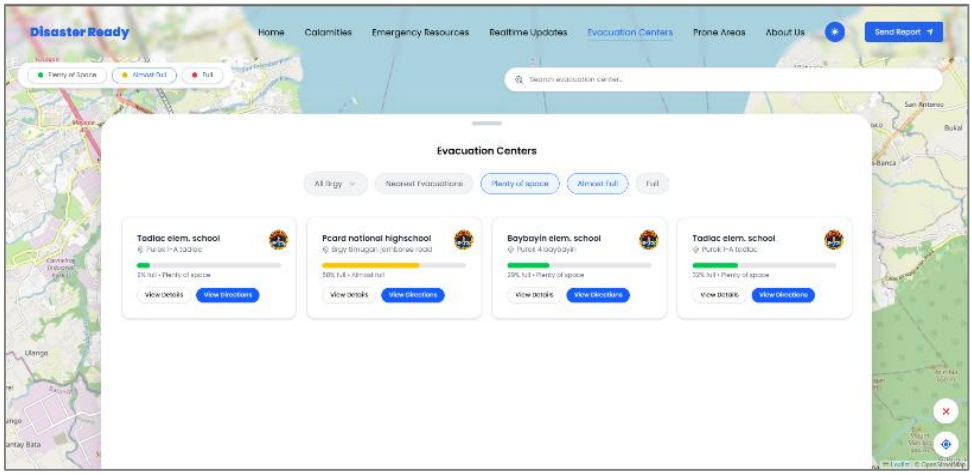


Figure 21. Evacuation Center List

Software Testing

Table 9. Unit Testing

Tested Module	Tested Function	Expected Output	Actual Output	Remarks
Homepage	Display home interface and navigation	The homepage loads successfully with all menus visible.	The homepage displayed completely within 3-5 seconds.	Passed
Donate Now	Submit the donation form	The system records valid donation entries and displays confirmation.	The donation was successfully recorded, and confirmation appeared within 4-7 seconds**.**	Passed
Incident Report	Submit an incident report	The report form validates input fields and saves data to the database.	Reports were validated and stored correctly in the database within 3-5 seconds**.**	Passed
Map	Visualize disaster-related locations	System displays the map with evacuation centers and important landmarks..	Map loads smoothly within 2–3 seconds and markers are correctly positioned.	Passed
Updates and News	Post and display advisories	The system saves new posts and displays them on the homepage.	Posts appeared immediately after submission within 2-3 seconds**.**	Passed
Relief Goods	Manage relief inventory	The system updates inventory records accurately after each entry.	Inventory updates reflected correctly in the database within 2-3 seconds**.**	Passed
Donation	See the history	Donations from users appear in the admin dashboard.	Donation records appeared and updated in real time within 3-5 seconds.	Passed
User Management	Add, edit, and deactivate user accounts	User accounts are validated and updated properly.	User records were updated successfully and reflected instantly within 4-5 seconds.	Passed
Disaster Risk Mapping	Load and edit risk data	The GIS map displays correct risk zones and allows updates.	The map loaded with accurate data and updates was saved successfully 3-5 seconds**.**	Passed
Barangay Contact Management	Add and view barangay contacts	Barangay contact details are saved and retrieved accurately.	Barangay data saved and displayed correctly within 3-5 seconds**.**	Passed

Table 10. Integration Testing of the System

Tested Case	Expected Output	Actual Output / Response Time	Remarks
Homepage	The system should properly display the homepage with navigation links to all modules and real-time disaster updates.	The homepage displayed all sections and links correctly, including real-time data and navigation menus. (3-5sec)	Passed
Donate Now	The donation form should accept valid inputs, record transactions, and reflect updates in the donation management module.	The donation form accepts and processes valid inputs, updating donation records instantly. (2-4 sec)	Passed
Incident Report	The system should allow users to submit incident reports and display them in the admin panel for verification.	Incident reports were submitted successfully and displayed in the admin panel in real time. (3-5 sec)	Passed
Notification	The system should send notifications to users for alerts and updates.	Notifications were sent and received instantly via the integrated messaging service. (1-3sec)	Passed
Updates and News	The system should display recent advisories, road updates, and community notices.	Updates and news are displayed correctly and refreshed in real time. (2-3 sec)	Passed
Relief Goods	The module should manage relief goods records and reflect distributions accurately.	Relief goods inventory and distributions were accurately updated in the system. (2-3 sec)	Passed
Donation	The admin panel should see the donation history data and reflect transactions from “Donate Now.”	Donation data synchronized correctly from the donor’s module. (4-6 sec)	Passed
User Management	The system should manage user accounts, including registration, activation, and deactivation.	User accounts were properly handled with accurate status updates. (4–8sec)	Passed
Disaster Risk Mapping	The system should visualize risk areas using GIS data and allow admin updates.	Disaster risk maps loaded successfully, displaying accurate hazard zones. (5–9 sec)	Passed
Barangay Contact Management	The module should display and update contact details for each barangay.	Barangay contact data is displayed and updated accurately in real time. (5-8 sec)	Passed

Table 11. Compatibility Testing

Browser	Web Page / Module	Performance	Accessibility	Best Practice	SEO
Chrome	Dashboard	97	88	100	100
Chrome	Evacuation Center	96	86	98	100
Chrome	Relief Goods	94	85	100	100
Chrome	Donation	95	84	100	100
Chrome	Barangay Contact Information	97	86	100	100
Chrome	Disaster Risk Mapping	96	84	98	100
Chrome	Updates and News	98	90	100	100
Chrome	User Management	97	87	100	100
Chrome	Account Setting	98	88	100	100
Firefox	Login	100	90	100	100
Firefox	Dashboard	96	87	100	100
Firefox	Evacuation Center	95	85	98	100
Firefox	Relief Goods	93	84	100	100
Firefox	Donation	94	83	100	100
Firefox	Barangay Contact Information	96	84	100	100
Firefox	Disaster Risk Mapping	95	83	98	100
Firefox	Updates and News	97	89	100	100
Firefox	User Management	96	85	100	100
Firefox	Account Setting	97	86	100	100
Microsoft Edge	Login	100	92	100	100
Microsoft Edge	Dashboard	97	88	100	100
Microsoft Edge	Evacuation Center	96	85	98	100
Microsoft Edge	Relief Goods	94	84	100	100
Microsoft Edge	Donation	94	83	100	100
Microsoft Edge	Barangay Contact Information	96	85	100	100
Microsoft Edge	Disaster Risk Mapping	95	83	98	100
Microsoft Edge	Updates and News	97	90	100	100
Microsoft Edge	User Management	96	87	100	100
Microsoft Edge	Account Setting	97	86	100	100

Table 12. Survey Questionnaire Interpretation

Survey Questionnaire	Mean	Interpretation
The web application is simple to use, easy to navigate, and user-friendly.	3.88	Strongly Agree
The design and layout of the Disaster Ready web app are clear, organized, and visually appealing.	3.75	Strongly Agree
The system allows users to quickly and accurately locate nearby evacuation centers.	3.64	Strongly Agree
The web app provides real-time updates and information about available resources (relief goods).	3.58	Strongly Agree
The web application functions efficiently with minimal loading time and errors.	3.55	Strongly Agree
The real-time update feature helps users stay informed during emergencies.	3.57	Strongly Agree
The system allows donors to easily contribute their donations (bank transfer, E-wallet, cash, or in-kind).	3.56	Strongly Agree
The application displays accurate and up-to-date information about affected barangays (flood or landslide)	3.53	Strongly Agree
The web app enhances disaster preparedness and coordination among users and responders	3.68	Strongly Agree
Overall, I am satisfied with the performance and usefulness of the Disaster Ready web application.	3.83	Strongly Agree
Overall Weighted Mean	**3.66**	**Strongly Agree**

The results in Table 12 show that the respondents strongly agreed with all statements, indicating a highly positive perception of the *Disaster Ready* web

application. With an overall high mean range (3.53–3.88), users found the system simple, user-friendly, and efficient in delivering its intended functions. The highest-rated items highlighted the application's ease of navigation (3.88) and overall user satisfaction (3.83), demonstrating that residents and administrators could interact with the platform seamlessly. Other features such as real-time updates, accurate information about affected barangays, and easy donation management, were also rated favorably, showing that users appreciated the system's ability to provide timely and relevant data during emergencies.

Moreover, these findings suggest that the Disaster Ready web application effectively bridges the communication gap between local authorities and residents by offering accessible and organized disaster-related information. The high ratings across all indicators reflect the platform's ability to enhance community preparedness, promote transparency, and encourage active participation in disaster response efforts. Additionally, the system's intuitive design and reliable functionality help users make quick and informed decisions, which is crucial during critical situations. Overall, the consistently strong ratings affirm that the Disaster Ready web application successfully meets user expectations and contributes significantly to building a more resilient and well-prepared community in the Municipality of Los Baños. These results highlight the system's potential for even broader implementation across neighboring municipalities. As such, the application can serve as a scalable model for local government units seeking to strengthen technology-driven disaster risk reduction strategies.

Testing using the ISO 25010 Model

Table 13. Summary of Software Evaluation Results

Criteria	Mean	Interpretation
Functional Suitability	4.50	Very High
Performance Efficiency	4.63	Excellent
Compatibility	4.83	Excellent
Usability	5	Excellent
Reliability	4.38	Very High
Security	4.50	Very High
Maintainability	4	Very High
Portability	4.75	Excellent
Overall Weighted Mean	**4.57**	**Excellent**

The software evaluation results indicate a highly favorable assessment across all ISO 25010 quality criteria, reflecting the system's overall effectiveness and reliability. The overall weighted mean of 4.57, interpreted as Excellent**,** signifies that users perceive the software as meeting or exceeding expectations in most performance areas. Among the criteria**,** Usability (5.00), Compatibility (4.83), Performance Efficiency (4.63), and Portability (4.75) received the highest ratings, each interpreted as Excellent. These results suggest that the system is intuitive, easy to use, works well across different platforms or environments, responds efficiently during operation, and can be transferred or operated across various devices without major issues. High usability is particularly notable, as it reflects that users can navigate the

software with minimal difficulty, complete tasks efficiently, and adopt the system quickly—a critical factor for a disaster preparedness application where rapid access to information can directly impact response during emergencies.

To further analyze the evaluation results, it is useful to examine why Usability achieved the highest rating and why Maintainability received the lowest, highlighting the strengths and areas for improvement in the system. The system's Usability received the highest rating due to its clear interfaces, intuitive navigation, and concise instructions across all modules, including real-time updates, evacuation routes, and incident reporting. This design ensures that users of varying technical skill levels can operate the application effectively, minimizing errors and delays in accessing critical information. High usability supports rapid adoption and contributes directly to user satisfaction and confidence, making the software highly effective in emergency situations where quick decision-making is essential. In contrast, Maintainability**,** rated slightly lower at 4.00, reflects the complexity of the system, which involves multiple modules, real-time GIS integration, and dynamic data updates. Updating, modifying, or improving the software requires careful coordination and technical expertise, which can make maintenance more challenging. Enhancing maintainability through modular design improvements and streamlined update procedures could increase long-term robustness and reduce the effort required for future system enhancements.

In contrast, criteria such as Functional Suitability (4.50), Security (4.50), Reliability (4.38), and Maintainability (4.00) received Very High interpretations. Maintainability, being the lowest at 4.00**,** remains within the Very High range

but indicates that updating, correcting, or improving the software may require more effort compared to other attributes. This is likely due to the system's complexity, which includes GIS integration, real-time data updates, and multiple interdependent modules. Maintaining such a system requires technical expertise and careful coordination, making it inherently more challenging than tasks like navigating the interface or using core features.

Overall, the evaluation demonstrates that the software is well-developed, user-friendly, and capable of delivering high-quality performance. Minor improvements in reliability and maintainability could further strengthen its robustness, but the consistently high ratings across most criteria indicate strong potential for long-term implementation, scalability, and integration with evolving municipal systems. The high marks in usability and compatibility suggest that the software is well-positioned to accommodate a growing user base and support effective disaster preparedness.

CHAPTER V

SUMMARY, CONCLUSION, AND RECOMMENDATIONS

Summary

The study of Disaster Ready: A Web-Based Application for Evacuation and Resource Locator During Natural Calamities in the Municipality of Los Baños, was developed to assist residents, barangay officials, and the municipal government in improving disaster preparedness, response, and communication. The system provides essential features such as navigation to nearby evacuation centers, real-time disaster updates, submission of emergency reports, and access to safety tips before, during, and after calamities.

The development of the system aimed to address issues of delayed information dissemination, lack of centralized data, and difficulty in locating safe routes during emergencies. The researchers utilized web technologies to ensure accessibility and responsiveness across devices. The inclusion of modules for administrators and residents enhanced communication and coordination, particularly during typhoons, floods, landslides, earthquakes, and volcanic eruptions.

An ISO-based system evaluation was conducted covering functional suitability, performance efficiency, compatibility, usability, reliability, and security. The results showed that users rated the system as Very High to Excellent in all categories, indicating that “Disaster Ready” effectively met its objectives and user expectations.

Conclusions

Based on the study's findings, the researchers conclude that the Disaster Ready web-based application is an effective and reliable platform for enhancing disaster preparedness and response in the Municipality of Los Baños. By centralizing communication among the municipal government, barangay sub-administrators, and residents, the system enables timely and organized dissemination of critical information during emergencies. The integrated navigation and incident reporting features support safer decision-making by helping users identify secure routes and locations, thereby strengthening coordinated disaster response. Overall, the implementation of Disaster Ready has important implications for local disaster management, as it supports evidence-based planning, improves community awareness, and contributes to greater resilience and preparedness against natural calamities.

Recommendations

In light of the conclusions, the following recommendations are proposed:

1. **System Enhancement.** Future developers may integrate additional features such as offline access, SMS-based notifications, and real-time weather API integration to ensure continuous functionality during power or internet interruptions.
2. **User Training and Awareness.** Regular training and information campaigns should be conducted among residents and barangay officials to maximize the use of the system's features.

3. **Wider Implementation.** It is recommended that the system be expanded to other municipalities or cities to enhance regional disaster management and information sharing.
4. **Further Research.** Future studies may focus on mobile app integration, artificial intelligence-based hazard prediction, and data analytics to enhance the accuracy and usability of disaster-related information.
5. **Implementation Feasibility.** For the long-term sustainability and effective operation of the system, it is recommended that the Municipal Disaster Risk Reduction and Management Office (DRRMO) be designated as the official system owner, with barangay officials acting as sub-administrators for localized management. The system should be hosted on a reliable municipal or cloud server with sufficient bandwidth and storage to accommodate real-time updates and multiple simultaneous users. Regular maintenance, updates, and technical support should be coordinated by the DRRMO IT personnel, with clear protocols for troubleshooting, backup, and system upgrades. By ensuring proper hosting, ownership, and maintenance responsibilities, the system can remain functional, accessible, and scalable, thereby providing long-term benefits for disaster preparedness and response within the Municipality of Los Baños and potentially other regions.

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APPENDIX A

Survey Questionnaire

DISASTER READY: A Web-based Application for Evacuation and Resource Locator During Natural Calamities in the Municipality of Los Baños

Good day! We are graduating students of LSPU–LB from the College of Computer Studies, currently enrolled in the Capstone Project course. One of the requirements of the course is to conduct research.

This survey aims to identify the user satisfaction level of Disaster Ready: A Web-based Application for Evacuation and Resource Locator During Natural Calamities in the Municipality of Los Baños. The system is designed to assist residents, responders, and donors by providing real-time information on evacuation centers, resources, and emergencies.

Rest assured that all the information gathered from this survey would be used solely for academic purposes and would remain confidential.

We appreciate your cooperation.

Instructions:

Please rate our system using the scale below:

4 = Strongly Agree
3 = Agree
2 = Disagree
1 = Strongly Disagree

Name: _____
Gender: _____
Addres: _____

No.	Question	Strongly Agree	Agree	Disagree	Strongly Disagree
1	The web application is simple to use, easy to navigate, and user-friendly.				
2	The design and layout of the Disaster Ready web app are clear, organized, and visually appealing.				
3	The system allows users to locate nearby evacuation centers quickly and accurately.				
4	The web app provides real-time updates and information about				

	available resources (relief goods).				
5	The web application functions efficiently with minimal loading time and errors.				
6	The real-time update feature helps users stay informed during emergencies.				
7	The system allows donors to easily contribute their donations (bank transfer, E-wallet, cash or in-kind).				
8	The application displays accurate and up-to-date information about affected barangays (flood or landslide)				
9	The web app enhances disaster preparedness and coordination among users and responders.				
10	Overall, I am satisfied with the performance and usefulness of the Disaster Ready web application.				

APPENDIX B

IT Specialist Survey Questionnaire

A. Functional Suitability

	Poor (1)	Fair (2)	High (3)	Very High (4)	Excellent (5)
The system provides the necessary features for disaster preparedness and response (e.g., evacuation routes, alerts, and resource locations).					
The features of the system perform accurately and meet their intended purposes.					
The system's functions are relevant and useful in addressing users' needs during calamities.					
The information displayed by the system is complete and reliable.					

B. Performance Efficiency

	Poor (1)	Fair (2)	High(3)	Very High(4)	Excellent (5)
The system loads quickly and responds efficiently to user actions.					
The application maintains good performance even with multiple users accessing it simultaneously.					
The system processes and delivers updates in a timely manner.					
The overall performance of the application meets your expectations.					

C. Compatibility

	Poor (1)	Fair (2)	High(3)	Very High(4)	Excellent (5)
The system runs properly on different devices (mobile phones, tablets, laptops, etc.).					
The application is compatible with various browsers and operating systems.					
The interface adjusts well to different screen sizes and orientations.					

D. Usability

	Poor (1)	Fair (2)	High (3)	Very High(4)	Excellent (5)
The interface is easy to navigate and user-friendly.					
The design and layout make it simple to understand the system's features.					
You can use the system efficiently without needing much instruction or help.					
The overall user experience is pleasant and intuitive.					

E. Reliability

	Poor (1)	Fair (2)	High (3)	Very High(4)	Excellent (5)
The system operates consistently without frequent errors or crashes.					
The application recovers quickly after interruptions (e.g., lost connection or power outage).					
Data entered in the system is saved securely and remains accessible.					
The system performs reliably under various conditions.					

F. Security

	Poor (1)	Fair (2)	High (3)	Very High(4)	Excellent (5)
The system ensures the protection and confidentiality of user data.					
Proper authentication (e.g., login, password) is required before accessing sensitive information.					
You feel confident that your information is safe within the system.					

G. Maintainability

	Poor (1)	Fair (2)	High(3)	Very High(4)	Excellent (5)
The web application's code structure is organized and easy to modify when updates are needed.					
The system allows developers to quickly diagnose and fix issues or bugs.					
The modules of the application are well-structured and reusable for future enhancements.					
The documentation provided (code comments, guides) is sufficient for maintaining the system.					

H. Portability

	Poor (1)	Fair (2)	High(3)	Very High(4)	Excellent (5)
The system performs consistently on various operating systems (Windows, Android, iOS).					
The web app adjusts effectively to different screen sizes					

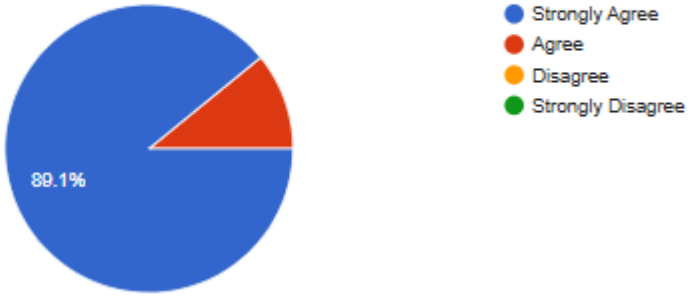
and resolutions.					
The application remains compatible and fully functional across different web browsers.					
The system can be migrated to alternative hosting platforms or servers with ease.					

APPENDIX C

Result of Post - Evaluation

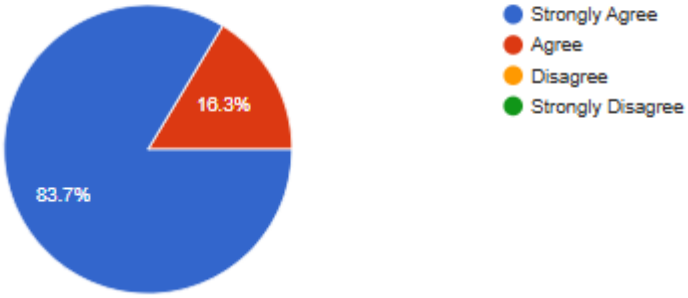
1. The web application is simple to use, easy to navigate, and user-friendly.

257 responses



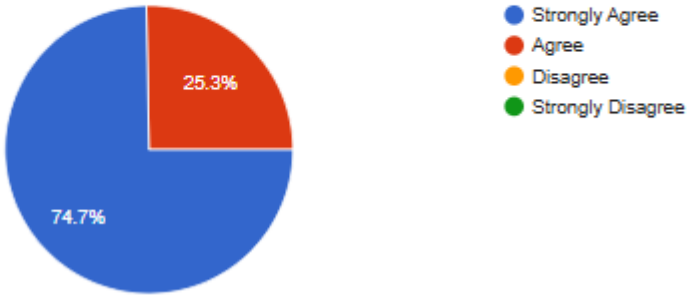
2. The design and layout of the Disaster Ready web app are clear, organized, and visually appealing.

257 responses



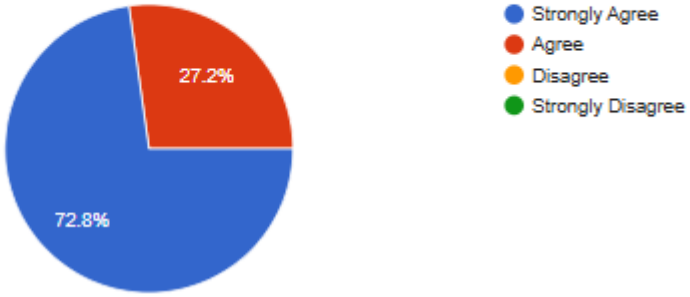
3. The system allows users to locate nearby evacuation centers quickly and accurately.

257 responses



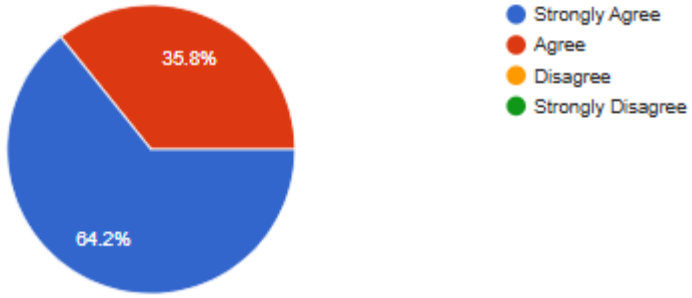
4. The web app provides real-time updates and information about available resources (relief goods).

257 responses



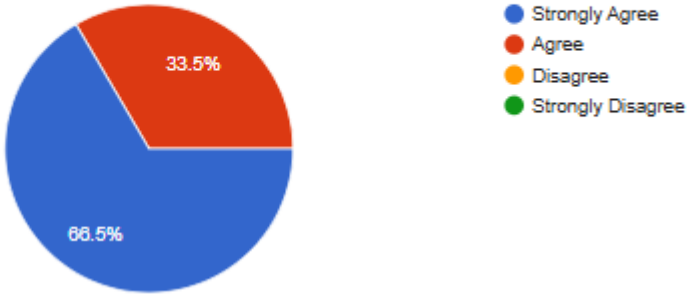
5. The web application functions efficiently with minimal loading time and errors.

257 responses



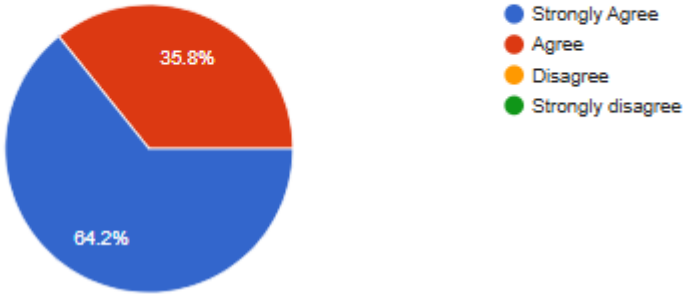
6. The real-time update feature helps users stay informed during emergencies.

257 responses



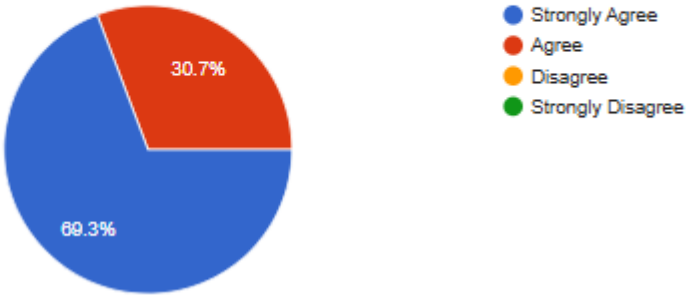
7. The system allows donors to easily contribute their donations (bank transfer, E-wallet, cash or in-kind).

257 responses



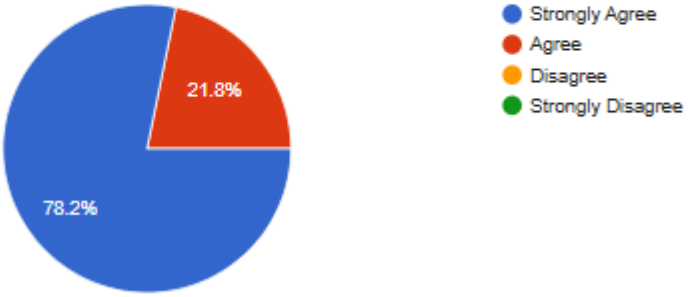
8. The application displays accurate and up-to-date information about affected barangays (flood or landslide)

257 responses



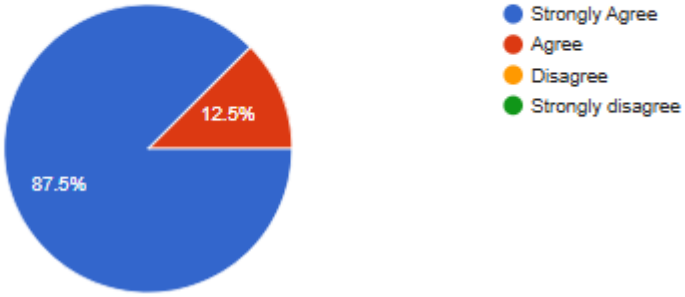
9. The web app enhances disaster preparedness and coordination among users and responders.

257 responses



10. Overall, I am satisfied with the performance and usefulness of the Disaster Ready web application.

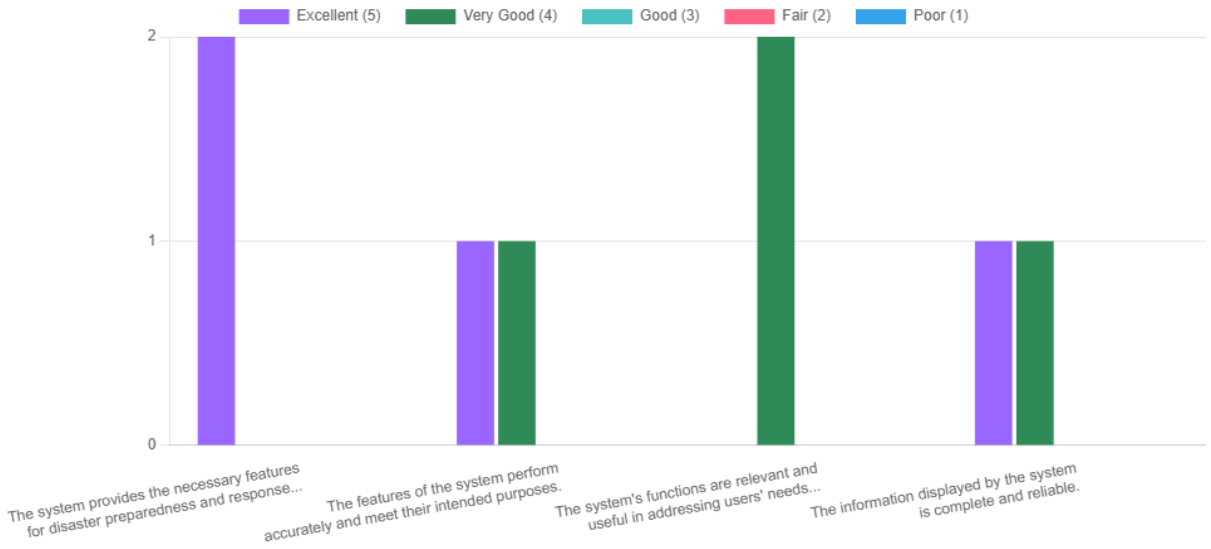
257 responses



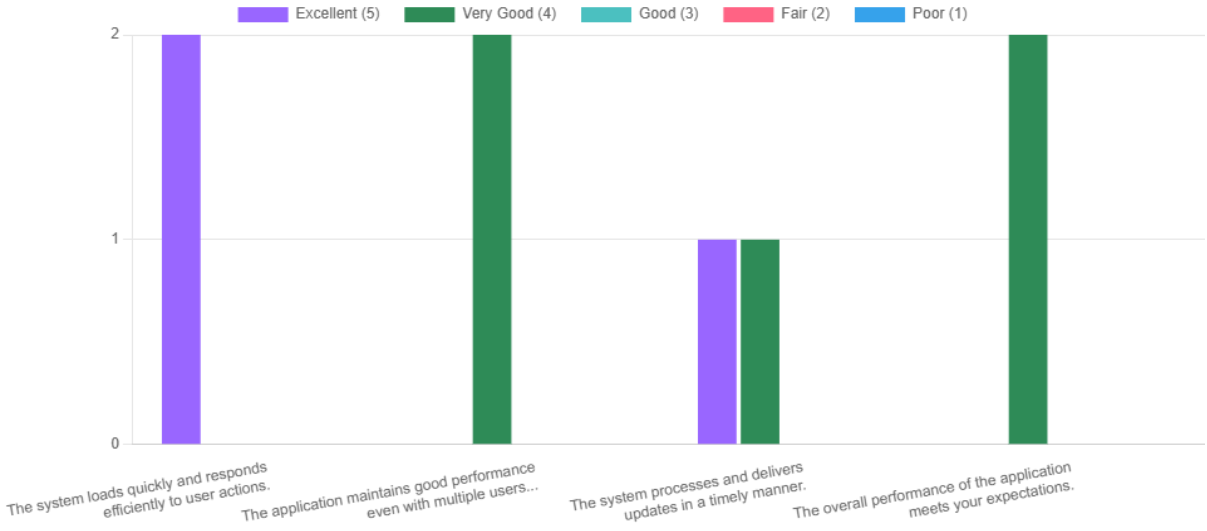
APPENDIX D

Result of ISO 25010 Evaluation

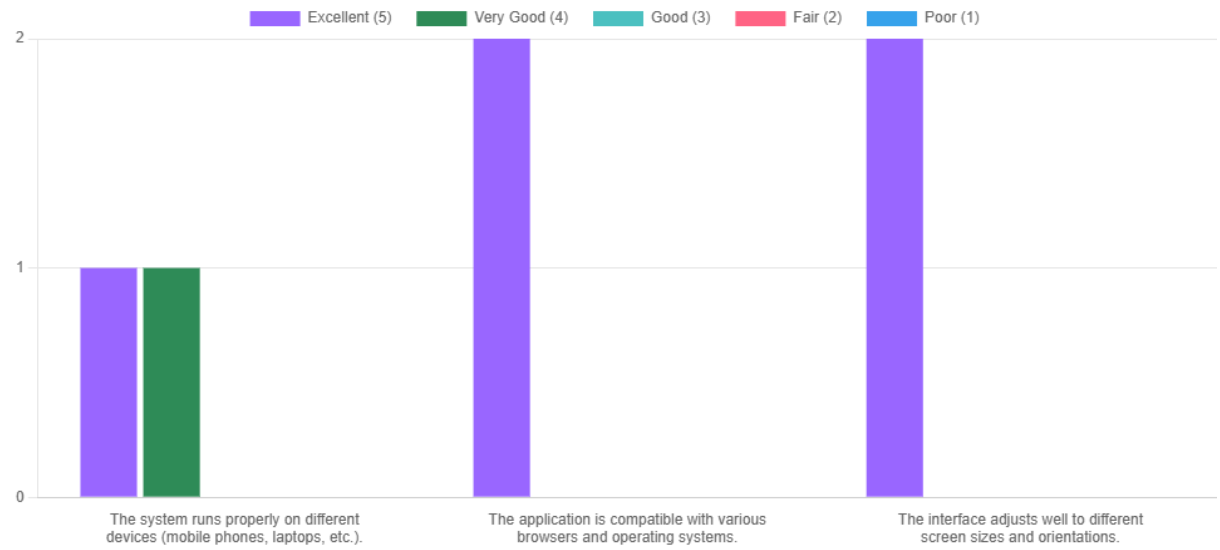
Functional Suitability



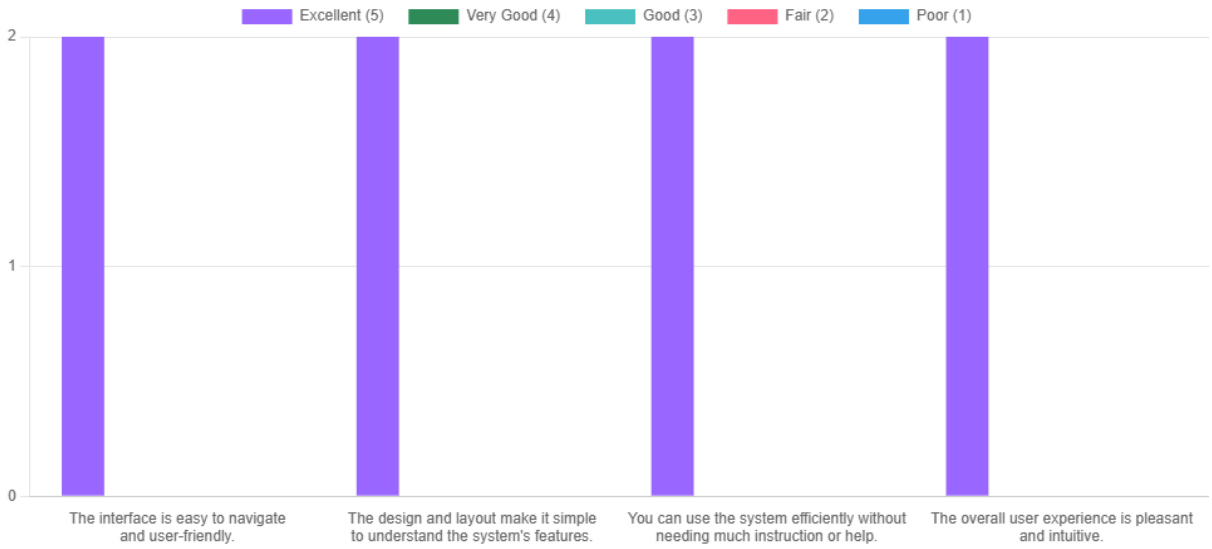
Performance Efficiency



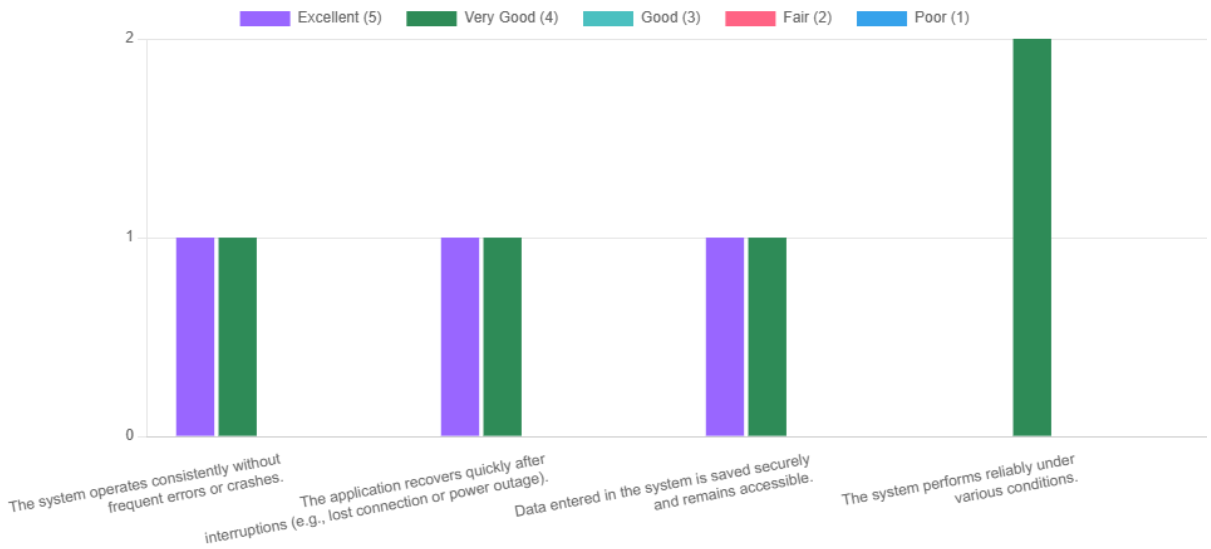
Compatibility



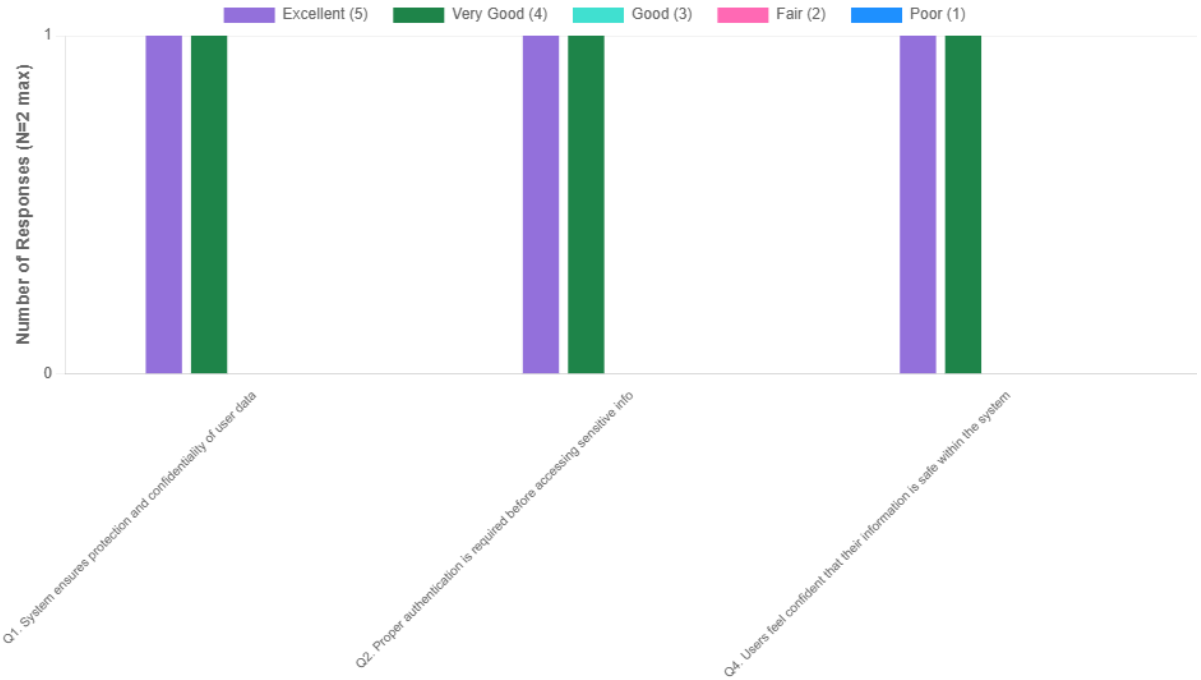
Usability



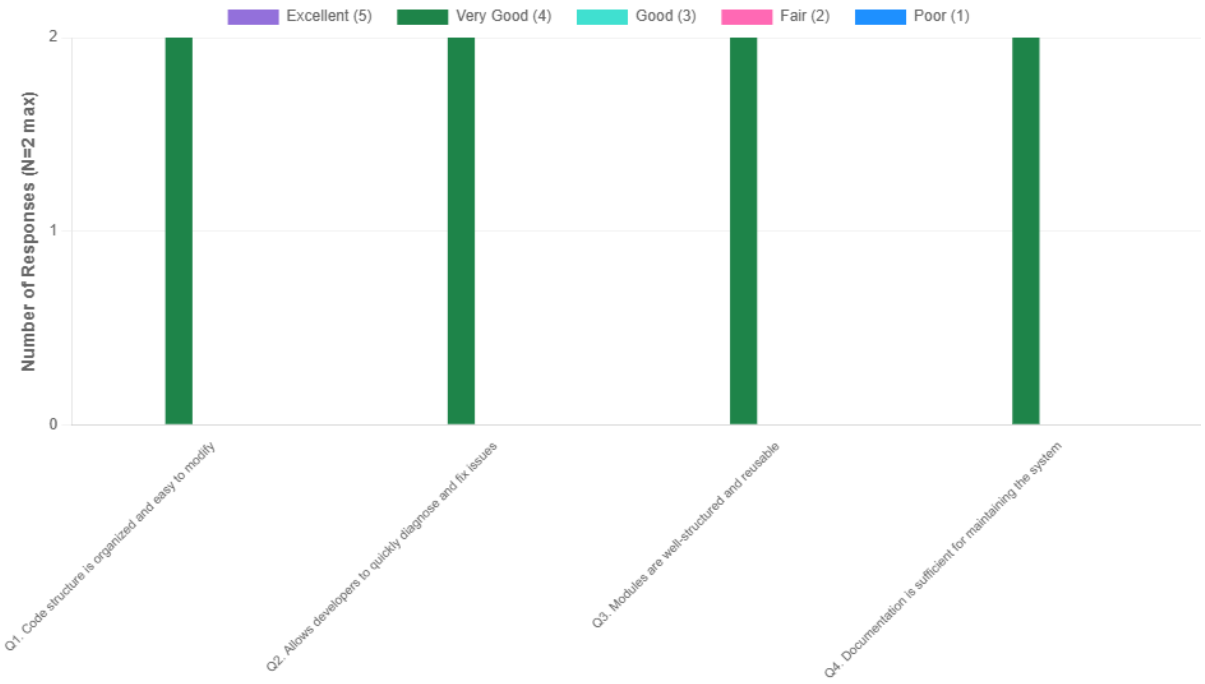
Reliability



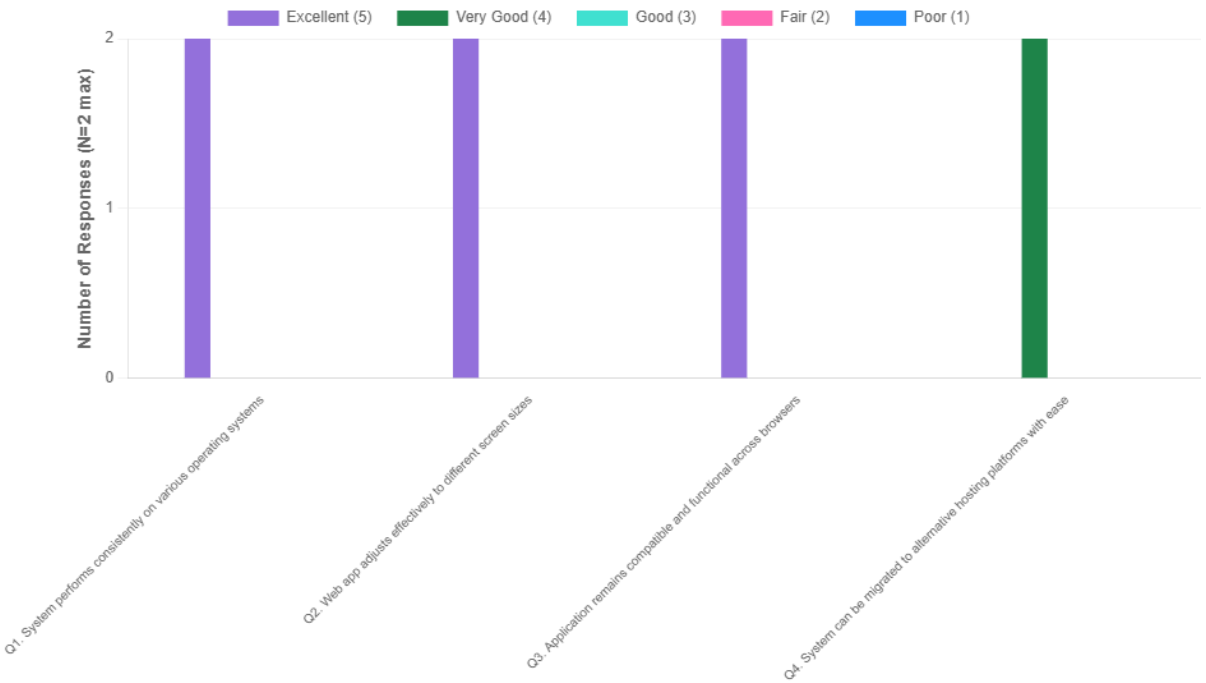
Security



Maintainability



Portability



APPENDIX E

Curriculum Vitae



Kylenefaye Jimeno

+639631482753

kfjimeno@gmail.com

Grove St. Brgy. Batong Malake, Los Baños, Laguna

CAREER OBJECTIVE

A graduating BSIT student from LSPU, specializing in Web and Mobile Application Development. Proven success in team-based collaborative projects, demonstrating strong skills in time management and effectively communicating technical requirements within a group setting. Eager to contribute creativity and technical proficiency to dynamic development teams.

PERSONAL INFORMATION

Age: 23

Sex: Female

Nationality: Filipino

Civil Status: Single

Languages: English, Filipino

SKILLS

- Teamwork
- Time Management
- Problem Solving
- Adaptability
- Communication Skill

EDUCATIONAL BACKGROUND

Laguna State Polytechnic University (LSPU)

Bachelor of Science in Information Technology

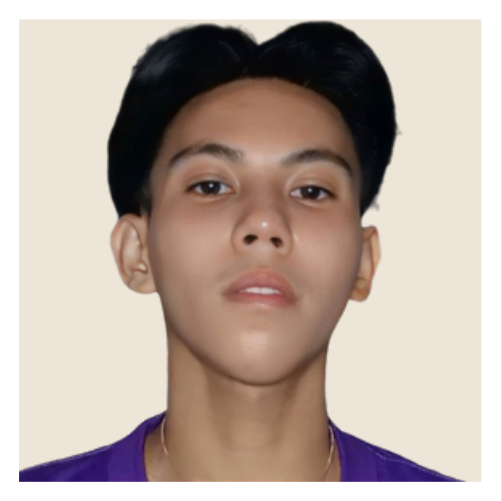
Web and Mobile Application Development

2022 – Present

Haluban National High School

General Academic Strand (GAS)

2018 - 2020



Harvey Adorco

Front End Dev

I create websites that are easy to use and visually appealing. I focus on building projects that are functional, user-friendly, and have a clean design.

Personal Information

Name: Harvey Adorco

Age: 21

Sex: Male

Nationality: Filipino

Civil Status: Single

Language: English, Filipino

Birthday: March 10 2004

Address: Purok 1-A Tadlak

Losbaños Laguna

Skills

. Programming Languages

- HTML
- CSS
- JavaScript
- TypeScript

Front-End Frameworks / Libraries

- React.js
- Next.js
- React Hook Form
- Zod
- TanStack Query (React Query)
- Tailwind CSS

Tools & Workflow

- Git
- GitHub
- VS Code
- REST API Integration
- Responsive Web Design
- UI/UX Basics
- Component-Based Development

Contact Information

Email: harveyadorco2020@gmail.com

Contact no. : 09104552659

Facebook: Harvey Adorco

Instagram: Its_Harvs

Github: Harvey102004

Education

Laguna State Polytechnic University (LSPU)

Bachelor of Science in Information Technology

Web and Mobile Application Development

2022 – Present

Lalakay Senior Highschool

2020 – 2022



Jared Abrera

+639197017084

jaredabrera44@gmail.com

Simon St. Barangay Mayondon, Los Baños, Laguna

CAREER OBJECTIVE

A motivated BSIT student specializing in backend development, aiming to apply technical skills in building reliable and secure systems. Currently contributing as the backend developer for the Disaster Ready capstone project, committed to delivering efficient solutions and gaining practical experience in real-world system development.

PERSONAL INFORMATION

Age: 21

Sex: Male

Nationality: Filipino

Civil Status: Single

Languages: English, Filipino

SKILLS

- Problem-Solving
- Attention to Detail
- Teamwork
- Time Management
- Analytical Thinking
- Adaptability

EDUCATIONAL BACKGROUND

Laguna State Polytechnic University (LSPU)

Bachelor of Science in Information Technology

Web and Mobile Application Development

2022 – Present

Luzonian Center of Excellence for Science and Technology

Technical-Vocational-Livelihood (TVL) – Mechatronics Strand

2020 – 2022

APPENDIX F

Publishable Paper (IEEE format)

DISASTER READY: A WEB-BASED APPLICATION FOR EVACUATION AND RESOURCE LOCATOR DURING CALAMITIES IN THE MUNICIPALITY OF LOS BAÑOS

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College of Computer Studies
Laguna State Polytechnic University
Laguna, Philippines
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Abstract - The study **"**Disaster Ready: A Web-Based Application for Evacuation and Resource Locator During Natural Calamities in the Municipality of Los Baños**"** was developed to improve disaster preparedness, communication, and response. The system provides residents and local authorities with a centralized platform for locating evacuation centers, submitting emergency reports, and accessing safety tips for typhoons, floods, landslides, earthquakes, and volcanic eruptions. Featuring real-time updates and web accessibility across devices, the application enhances coordination during disasters. An ISO-based evaluation of functional suitability, performance, compatibility, usability, reliability, and security rated the system from Very Good to Excellent, demonstrating its effectiveness. Disaster Ready offers a user-friendly and reliable solution for disaster management, with recommendations for offline features, real-time weather monitoring, and improved mobile responsiveness.

Keywords: *Disaster Management, Web-Based Application, Evacuation Locator, Community Preparedness, Los Baños*

1. INTRODUCTION

PROJECT CONTEXT

Disasters, both natural and man-made, pose a significant threat to communities, often leading to loss of lives and property. The Municipality of Los Baños, with a total population of 117,030, is particularly vulnerable to various natural calamities such as typhoons, floods, and earthquakes. Given this, there is a pressing need for a more efficient and technology-driven solution to enhance disaster preparedness and response. Traditional methods of emergency reporting and evacuation coordination often result in delays, misinformation, and inefficiencies, which can worsen the impact of disasters on affected communities.

According to Ruth Anna, a disaster is an unpredictably occurring event, such as a sudden event, an accident, or natural disasters that cause significant damage or death. The term "Disaster" is used loosely to refer to any incident, manmade accident, or natural occurrence that might affect the project's operation in any way. We are not evaluating the impact of the next disaster. There are two kinds of natural and manmade disasters. Disasters have affected not only human beings but also animals and all life on Earth. Disaster causes massive building damage or economic loss. It has a profoundly negative impact on the country's economy. (Ruth Anna, 2009)

Evacuation is fundamental to saving lives during natural disasters. People in the Philippines may be endangered if they do not know the proper way to evacuate during disasters, especially during stampedes in Local Government Units (Atacador, Iglesias, & Beron, 2019). Traffic congestion, especially during typhoons, can trap people. The average number of typhoons per year in the Philippines makes it difficult to handle. The primary function of evacuation is to prevent damage and save people in safe zones. Early warnings and emergency announcements can reduce fatalities (Yan, Liu & Song, 2018).

Project Objectives

The main objectives of this study are:

1. Develop a Web-based Application to inform residents about disaster preparedness and partner with local officials for disaster information.

2. Provide real-time information for available evacuation centers and alternative routes using GIS technology, especially during live system operations.

3. Conduct system testing using the TAM and ISO25010 models to ensure system quality, reliability, and user acceptance.

2. RELATED LITERATURE

Disasters

A disaster is an unpredictable event, such as an accident or natural calamity, that causes significant damage or loss of life. The term refers to any manmade or natural event that disrupts normal operations and affects communities. There are two main types of disasters: natural and manmade. Disasters can impact humans, animals, and the environment, often resulting in economic loss and infrastructure damage (Ruth Anna, 2009).

Evacuation Centers

Evacuation centers are temporary shelters for people displaced by disasters. These safe zones provide essential needs such as food, water, and clothing (Onorati et al., 2014). In Los Baños, Laguna, several barangays—including Tadlac, Lalakay, Baybayin, Mayondon, Malinta, and Bambang—are prone to flooding due to proximity to Laguna Lake (Catapang, 2016; Grefaldo, 2016). The local government designates schools and multipurpose centers, such as Los Baños National High School and Lopez Elementary School, as evacuation sites (MDRRMO, 2016). Knowing the capacity of each site is vital for the LGU to prevent overcrowding and ensure organized evacuations (GovPilot, 2021).

Web-Based Application

Web-based applications operate through browsers, offering real-time access, scalability, and convenience without installation. They are widely used in fields like health, education, and disaster management (Alshamrani & Bahattab, 2021). As stated by Dahiya and Bhatia (2020), these apps are accessible anytime and anywhere, which is crucial during emergencies. Integrating geolocation tools, such as Google Maps API, allows real-time tracking of evacuation centers and affected zones (Nguyen et al., 2019). A user-friendly interface and secure data handling are vital in ensuring accessibility, reliability, and user trust (Nielsen, 2020; Owens et al., 2021).

Web-Based Geographic Information Systems (Web GIS)

Web GIS provides interactive visualization and analysis of spatial data directly through a browser, improving accessibility and decision-making (Zhang et al., 2018). In disaster management, it helps identify hazard-prone areas, map evacuation centers, and monitor resource distribution. By enhancing situational awareness, Web GIS supports quick and informed decisions (Goodchild & Glennon, 2010). Popular technologies like Leaflet.js and ArcGIS Online enable real-time mapping and updates (Esri, 2019). Because these systems are mobile-friendly, residents can easily access location-based disaster information even in remote areas (Chen et al., 2021).

Conceptual Framework

The process starts with the User, who can submit emergency reports and access donation features through the application. The data is processed by the web-based backend, built with PHP and MySQL, ensuring accuracy and reliability. Two types of administrators—Sub-Admin and Super-Admin—manage the system. The Sub-Admin validates reports and updates information, while the Super-Admin oversees system operations, manages donations, and ensures all data is accurate and up to date.

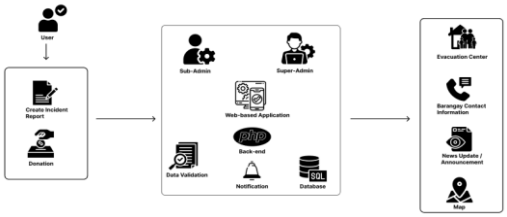


Figure 1. Conceptual Framework

3. METHODOLOGY

The Disaster Ready web-based application is organized into modules accessible by three roles: Super Admin and Sub Admin, all starting through the Login Module, which supports account creation, password recovery, and login. Users can access features such as Calamity Information, Emergency Resources, Updates and News, Incident Reporting, Disaster Risk Mapping, and Prone Areas. They can also donate, learn about the app, and manage their accounts. Super Admins have full control—managing evacuation centers, maps, users, donations, and reports—while Sub Admins handle updates and validations but have limited access. All modules connect to a central database, ensuring synchronized, secure, and real-time data for efficient disaster response and management. This structure allows seamless coordination between administrators and residents, enhancing communication and decision-making during emergencies. Overall, the system ensures that critical disaster information is always accessible, organized, and up to date. The design promotes transparency and accountability by ensuring that every update, donation, and report is properly tracked within the system. Its centralized platform also reduces manual processes, helping responders act faster during

urgent situations. Ultimately, the system strengthens community preparedness by providing a reliable digital tool for both everyday monitoring and crisis management.



Figure 2. System Architecture

3.2 Project Locale

This study was conducted in the Municipality of Los Baños, Laguna, a town frequently affected by natural disasters such as typhoons, floods, and earthquakes due to its geographical location and natural surroundings. The area was chosen as the project locale to develop "Disaster Ready," a web-based application designed to help residents and local authorities manage evacuation efforts and access essential resources during emergencies. The system aims to enhance real-time communication, improve accessibility to emergency data, and strengthen coordination between barangay officials and the community.

Project Development

The development of software applications typically follows a structured series of steps, commonly known as the Software Development Life Cycle (SDLC). This cycle is designed to guide developers through the necessary processes to build and deliver functional software solutions effectively. The SDLC typically encompasses key stages, including Planning**,** Requirements**,** Design**,** Development**,** Testing**,** Deployment**,** and Maintenance**.** These stages are often divided into six to eight steps, depending on the complexity and scale of the project.

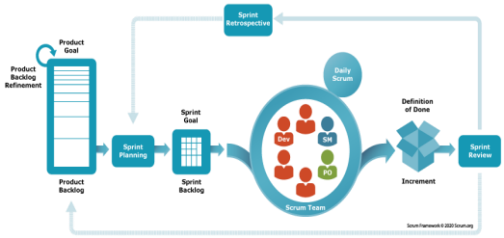


Figure 3. Agile Scrum

Sampling Design

Purposive sampling will be used to select residents of Los Baños, barangay officials, and emergency responders who are directly involved in disaster preparedness and response. This approach ensures that data collected reflects the actual needs and experiences of users most affected by calamities, helping to enhance the usability, reliability, and effectiveness of the Disaster Ready application in real-life disaster situations.

4. RESULT AND DISCUSSION

1. Web-based Application to inform residents about disaster preparedness and partner with local officials for disaster information.

The Disaster Ready web-based application effectively achieved its objectives by providing a centralized platform that enhances disaster preparedness and communication within the Municipality of Los Baños. The Home Page serves as the main hub for updates on weather, news, and announcements, promoting awareness and readiness among residents. The Calamity Page offers safety tips and guidelines before, during, and after disasters to help minimize risks and ensure proper response. The Emergency Resources Module enables users to contact barangay officials easily, improving coordination during emergencies. Through the Real-Time Updates Module**,** residents can view active incidents and resources via an interactive Leaflet map for better situational awareness. The Evacuation Center Module displays nearby shelters with real-time capacity indicators, guiding residents to safe locations while assisting authorities in managing evacuees. Lastly, the Prone Areas Module identifies flood- and landslide-prone zones such as Malinta, Mayondon, Bayog, and Lalakay, supporting proactive planning and risk reduction. Overall, the system's features collectively strengthen community preparedness, information accessibility, and coordinated disaster response in Los Baños.

2. **Real-time information for available evacuation centers and alternative routes**

using GIS technology, especially during live system operations

For Objective No. 2**,** the Disaster Ready web-based application successfully provides real-time information on available evacuation centers and alternative routes using GIS and Leaflet integration**.** The system displays evacuation centers across Los Baños with color-coded indicators showing occupancy levels—green for available, yellow for almost full, and red for full—allowing residents to make quick and informed decisions during emergencies. It also generates the most efficient evacuation routes from the user's current location to their chosen center, offering step-by-step navigation for safer travel. Overall, this feature enhances situational awareness, reduces confusion, and ensures efficient and well-coordinated evacuations during disaster situations.

Table 10. Summary of Software Evaluation Results

Criteria	Mean	Interpretation
Functional Suitability	4.75	Excellent
Performance Efficiency	4.25	Very Good
Compatibility	5	Excellent
Usability	4.88	Excellent
Reliability	4.13	Very Good
Security	4.13	Very Good
Maintainability	4	Very Good
Portability	4.75	Excellent
Overall Weighted Mean	4.52	Excellent

The results in Table 10 show that the Disaster Ready web-based application performed strongly, achieving an overall weighted mean of 4.52, interpreted as Excellent. Compatibility, Usability, and Functional Suitability received the highest ratings, indicating that the system works smoothly across devices, is easy to use, and effectively delivers its core functions such as real-time evacuation center information and alerts. Although Performance Efficiency, Reliability, and Security were rated Very Good, they still suggest minor areas for improvement to further enhance speed, stability, and data protection. Overall, the findings confirm that the application is a reliable and effective tool for disaster preparedness and response in Los Baños, significantly supporting community safety and coordination during emergencies. These results demonstrate that the system is well-received by users and is suitable for real-world deployment.

5. CONCLUSION AND RECOMMENDATION

Based on the findings of the study, the researchers conclude that the Disaster Ready web-based application is an effective and reliable tool for promoting disaster awareness, preparedness, and quick response within the Municipality of Los Baños. The system successfully centralized communication between the municipality, barangay sub-admins, and residents, enabling faster and more organized dissemination of disaster-related information.

Furthermore, the navigation and reporting features have proven to be valuable in identifying safe routes and locations during emergencies. The evaluation results confirmed that the system is functionally suitable, efficient, compatible, user-friendly, and secure. Thus, the implementation of *Disaster Ready* can significantly improve community resilience and coordination during natural disasters.

Recommendations

In light of the conclusions, the following recommendations are proposed:

1. **System Enhancement:** Future developers may integrate additional features such as offline access, SMS-based notifications, and real-time weather API integration to ensure continuous functionality during power or internet interruptions.
2. **User Training and Awareness:** Regular training and information campaigns should be conducted among residents and barangay officials to maximize the use of the system's features.
3. **Wider Implementation:** It is recommended that the system be expanded to other municipalities or cities to enhance regional disaster management and information sharing.
4. **Further Research:** Future studies may focus on mobile app integration, artificial intelligence-based hazard prediction, and data analytics to enhance the

accuracy and usability of disaster-related information.

ACKNOWLEDGEMENT

We want to express our deepest gratitude to all the people who made the completion of this study, “Disaster Ready: A Web-based Application for Evacuation and Resource Locator During Natural Calamities in the Municipality of Los Baños,” possible. We are sincerely thankful to our beloved parents for their endless love, encouragement, and support, whose understanding and sacrifices gave us the strength to finish this journey.

We also wish to thank our adviser, mentors, and panels for sharing their knowledge and guidance, as their valuable insights and feedback helped us improve and refine this research. Our appreciation extends to the local government units, barangay officials, and residents of Los Baños who willingly shared information and extended their support, making this study more meaningful and relevant to the community. We are equally grateful to the organizations and institutions that provided data, resources, and technical assistance, which were essential in completing this work.

Above all, we thank Almighty God for giving us wisdom, patience, and perseverance, and for guiding us every step of the way.

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APPENDIX G

Turnitin Similarity Report

DISASTER - REVISED MANUSCRIPT 2026 (2).pdf

Classroom #6

Academic Resilience II

Institute of Technology, Nirma University

Document Details

Submission ID

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134,943 Characters

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Grammarly Writing Assistant Report

CHAPTER 1-2 GRAMMARLY REPORT

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by Ronah joy Ramirez

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